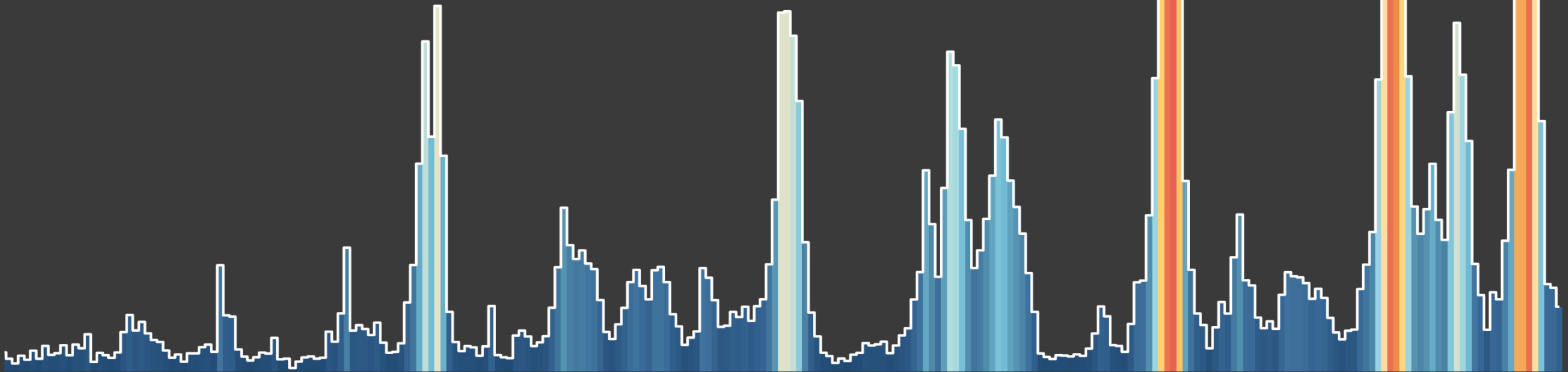


Why is biodiversity important for our health?

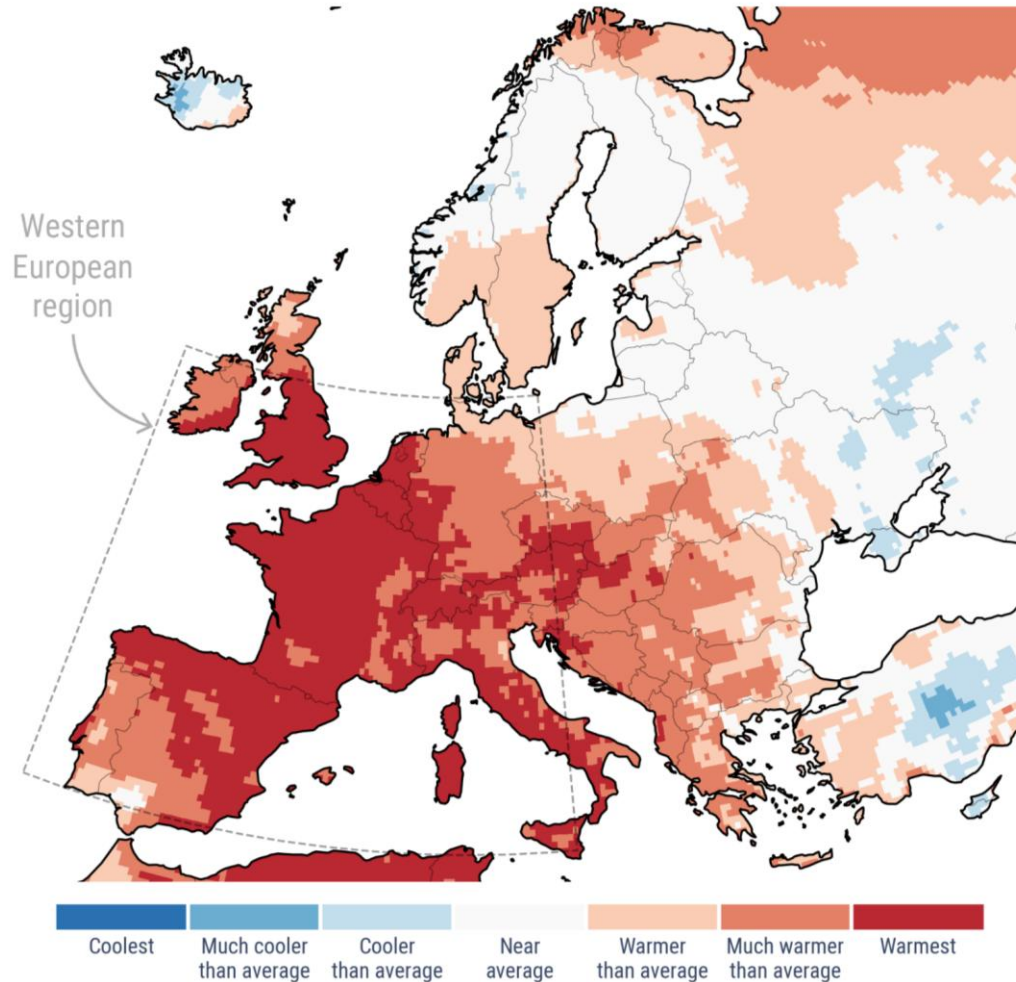


Dr. Rory Gibb

Royal Society University Research Fellow | People & Nature Lab



Anomalies and extremes in surface air temperature for June–August 2026



*“August was the
world’s joint hottest
month on record”*

(ECMWF/Copernicus)

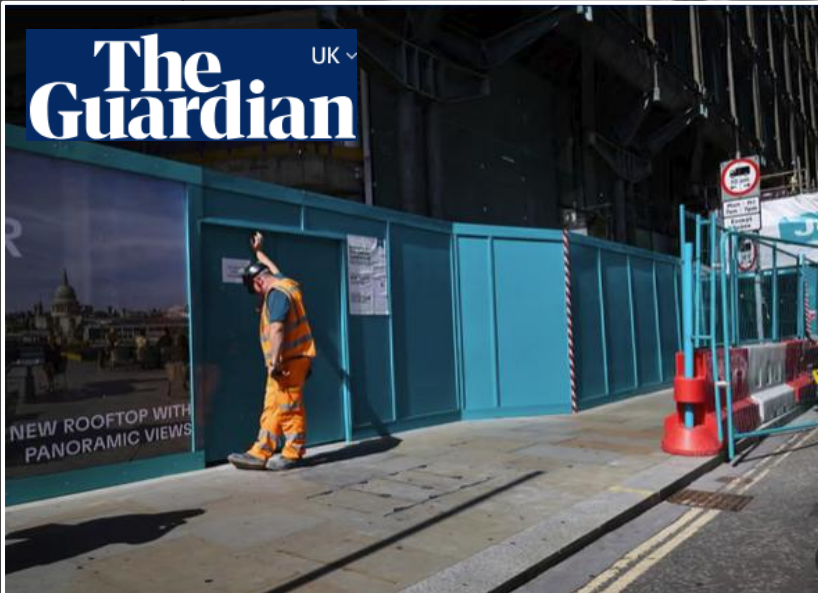
<https://climate.copernicus.eu/copernicus-august-was-worlds-joint-hottest-month-record-pushing-global-temperatures-back-above>

Power outages in France as Europe bakes in record heat

France and UK declare record-breaking temperatures as WHO says Europe heatwave putting people's 'health at risk'.

 Listen (6 mins)  Save  Share

 Add Al Jazeera on Google 



Extreme heat

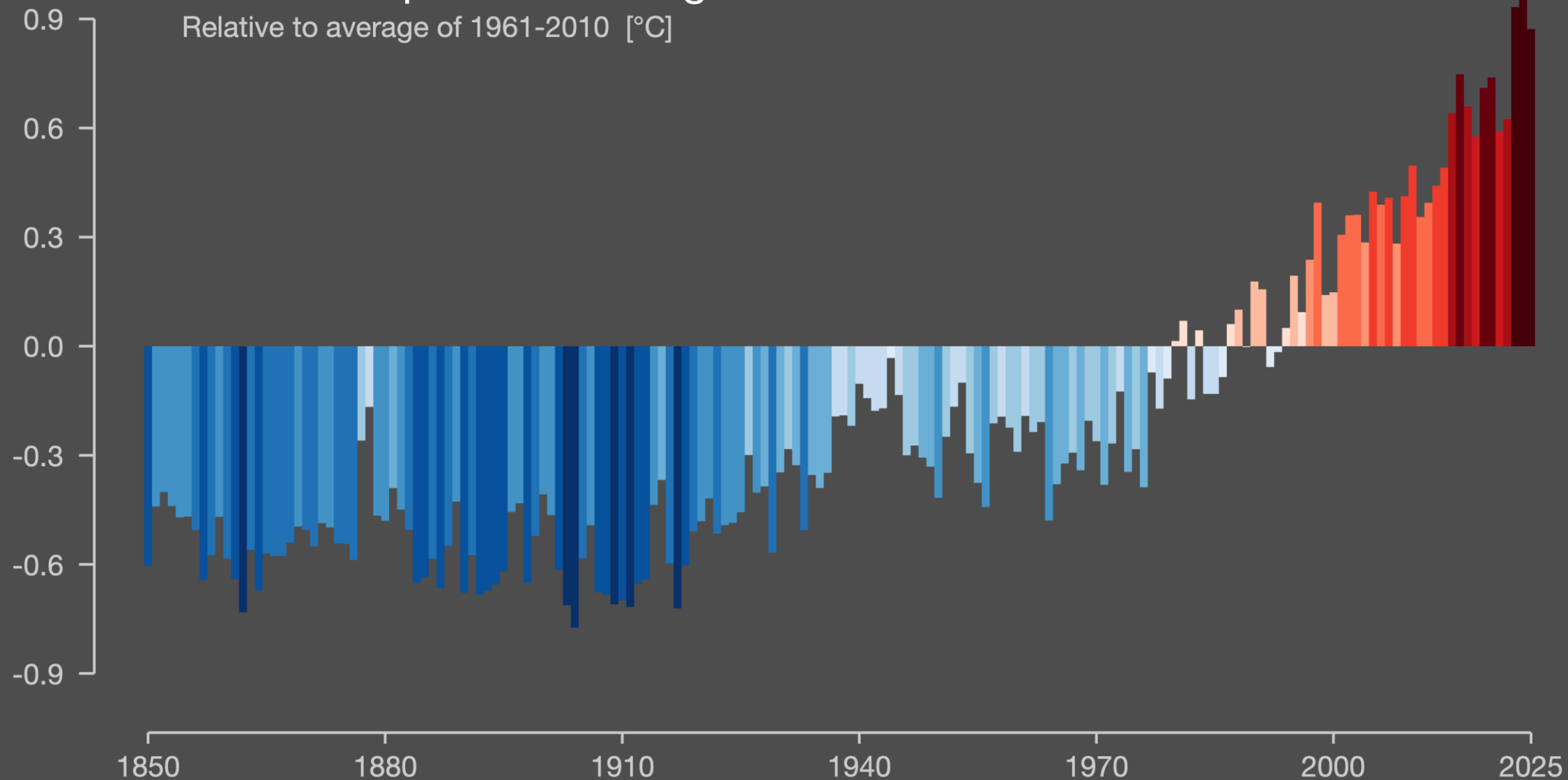
'It's dangerous': how schools, care homes and other UK workplaces are coping in searing heat

From bus drivers struggling to focus to those hauling scaffolding under a blazing sun, workers say risks must be taken seriously



Global temperature change

Relative to average of 1961-2010 [°C]



<https://showyourstripes.info/>

From global heating to global 'weirding'





Climate change is accelerating the biodiversity crisis

Environment Climate crisis **Wildlife** Energy Pollution

The Guardian UK



The age of extinction

Falling birds and dehydrated hedgehogs: heatwave takes its toll on UK wildlife

Animals and insects have been casualties of recent extreme heat, but the true impact will only be known over time, experts say

AF AMAZON FRONTLINES

STORIES & NEWS

THE TIPPING POINT: IS THE AMAZON RAINFOREST APPROACHING A POINT OF NO RETURN?

EARTH · ORG

BIODIVERSITY

Redistributing Life: How Climate Change Is Redrawing the Map of Species Migration

BY CLAUDIA DE LUCA
MAY 19TH 2026 | 7 MINS

EARTH.ORG IS POWERED BY OVER 150 CONTRIBUTING WRITERS



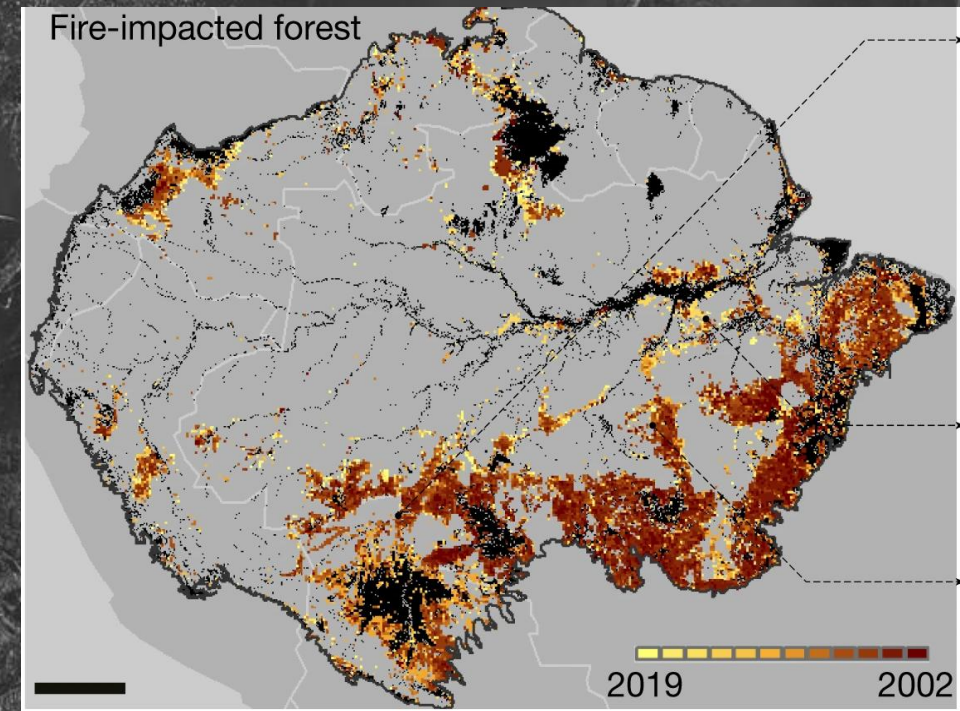
The biodiversity crisis is accelerating the climate crisis

The vicious cycle of climate change, deforestation, and fire in the Amazon



As long as deforestation continues in the Amazon, so will fire. And climate change is making it worse.

Update by [Sarah Ruiz](#), [Manoela Machado](#) • July 7, 2022



Feng et al, 2021, *Nature*

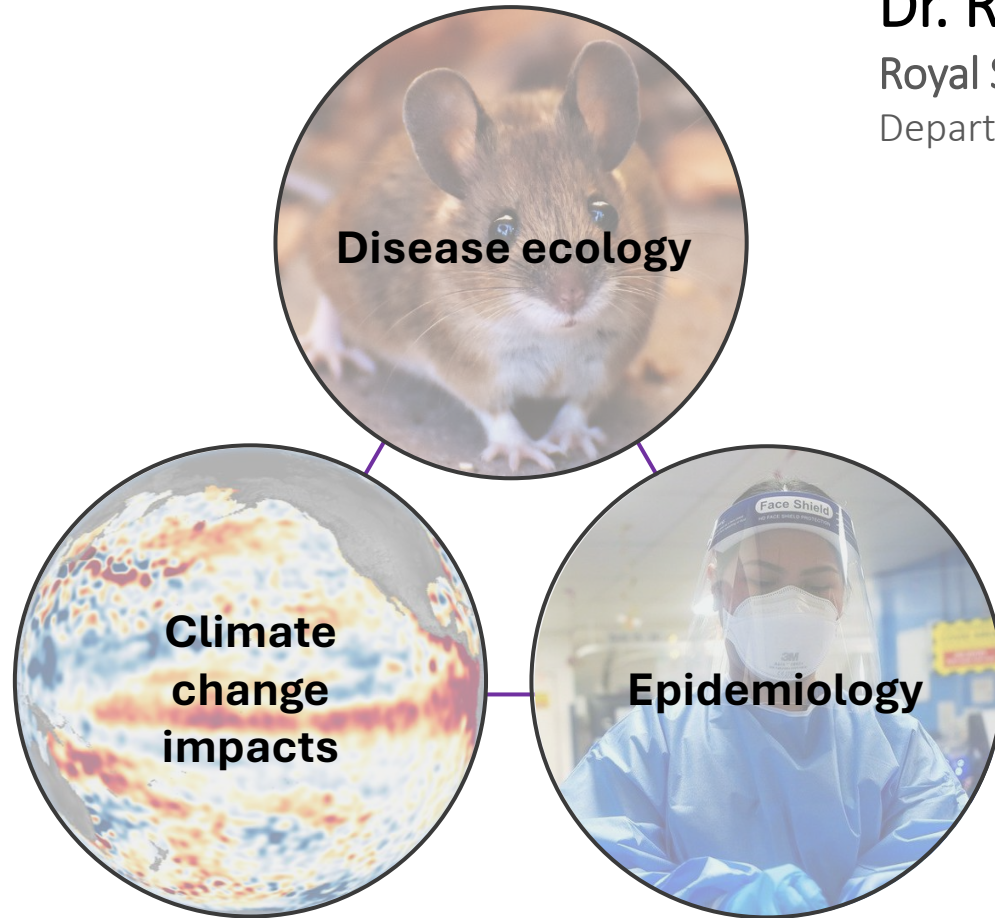
Why does all of this matter for our health?

Who am I?

Dr. Rory Gibb

Royal Society University Research Fellow

Department of Genetics, Evolution & Environment, Faculty of Life Sciences





Rory Gibb


Department of Genetics, Evolution & Environment, [University College London](#)
Verified email at ucl.ac.uk - [Homepage](#)
[Disease ecology](#) [zoonotic/vector-borne disea...](#) [climate change](#)
[epidemiology](#)

☐

TITLE




CITED BY

YEAR

☐

Zoonotic host diversity increases in human-dominated ecosystems

R Gibb, DW Redding, KQ Chin, CA Donnelly, TM Blackburn, T Newbold, ...
Nature 584 (7821), 398-402

996

2020

☐

Emerging opportunities and challenges for passive acoustics in ecological assessment and monitoring

R Gibb, E Browning, P Glover-Kapfer, KE Jones
Methods in Ecology and Evolution 10 (2), 169-185

780

2019

☐

Bat detective—Deep learning tools for bat acoustic signal detection

O Mac Aodha, R Gibb, KE Barlow, E Browning, M Firman, R Freeman, ...
PLoS Computational Biology 14 (3), e1005995

295

2018



<https://scholar.google.com/citations?user=ERU8djAAAAAJ&hl=en>

rory.gibb.14@ucl.ac.uk

Who am I?

Deputy director, MSc Ecology, Climate Change & Health

Ecology, Climate Change and Health MSc

Build multidisciplinary expertise of the impacts of global change on human and environmental health, and the data science skillset to develop operational solutions – on this applied one-year MSc.



Outline of this morning

Talk: Biodiversity and health
(9:15-10:15)

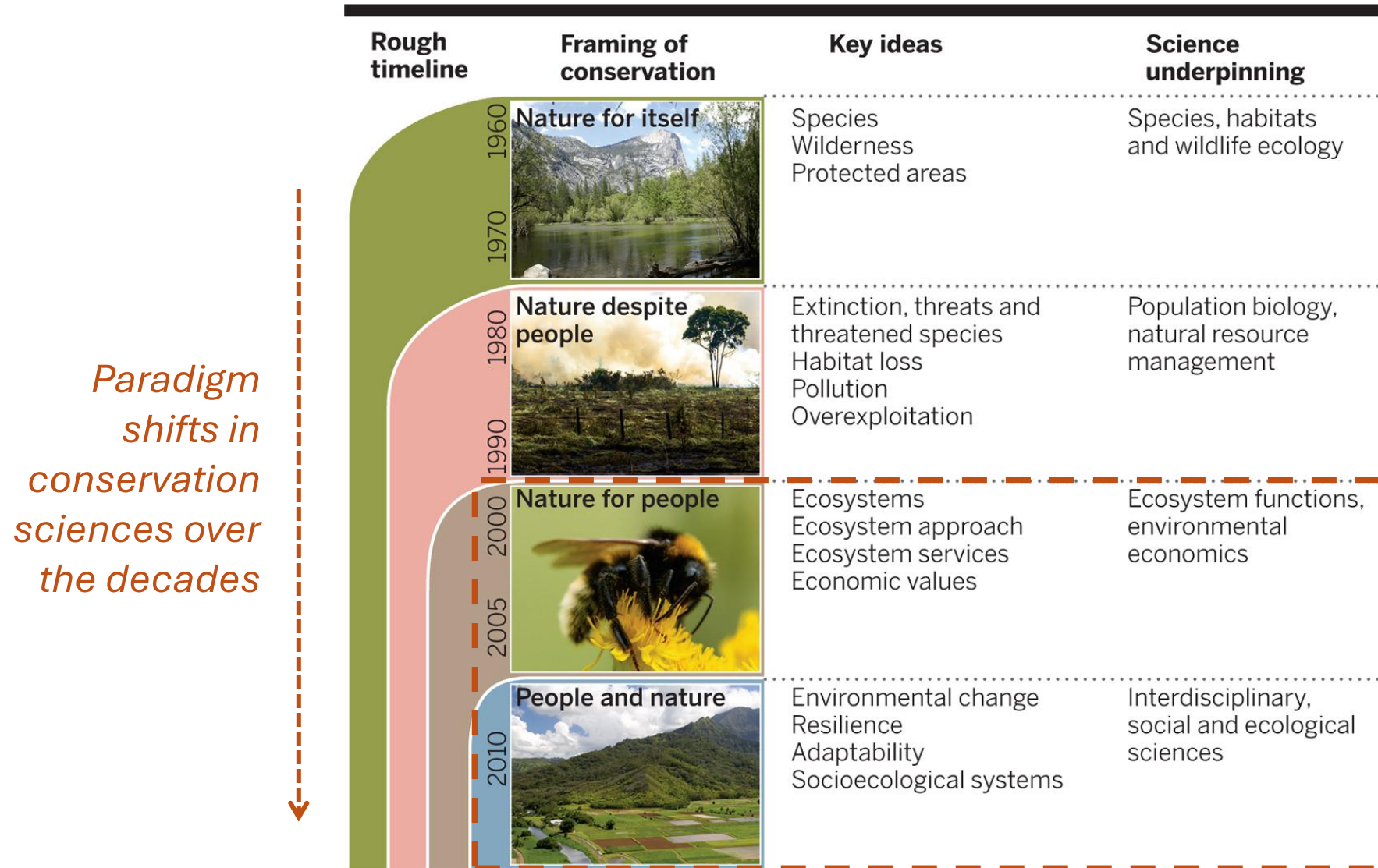
Practical workshop: Vector-borne disease emergence
(10:30-12:30)

Talk objectives

How do biodiversity and ecosystems contribute to human health and wellbeing?

How can we measure how ecological degradation is impacting health?

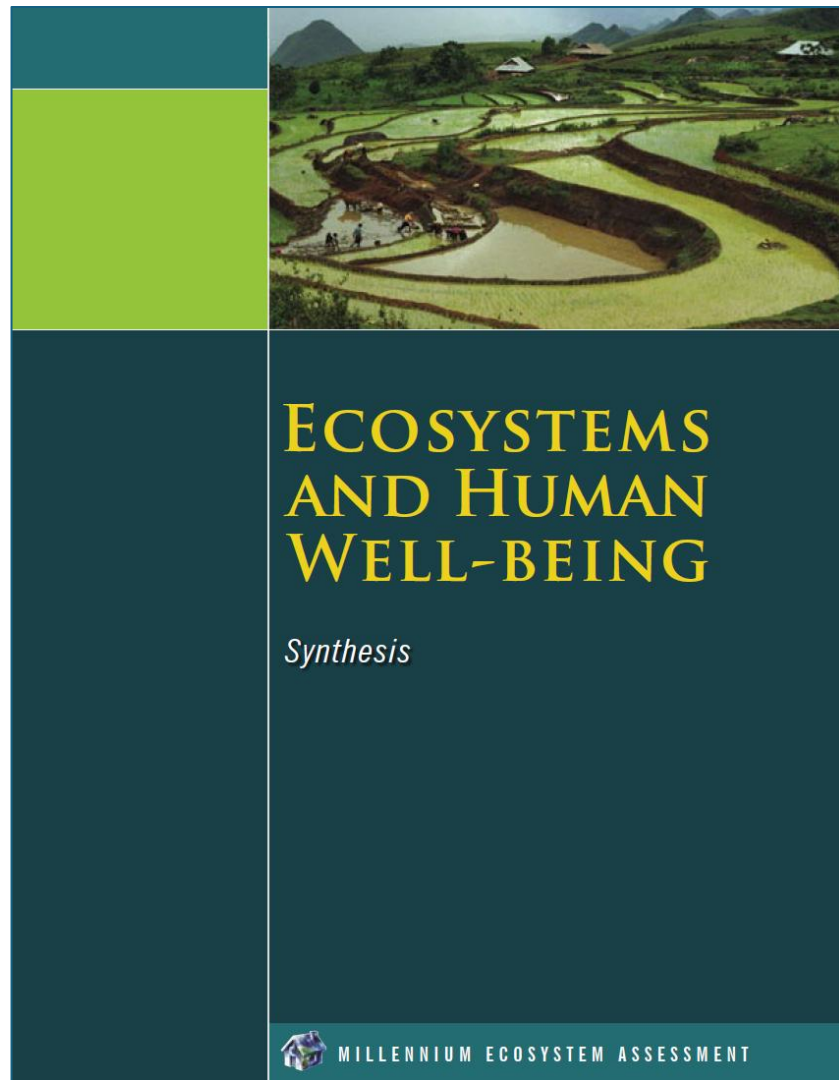
Changing views of biodiversity conservation through time



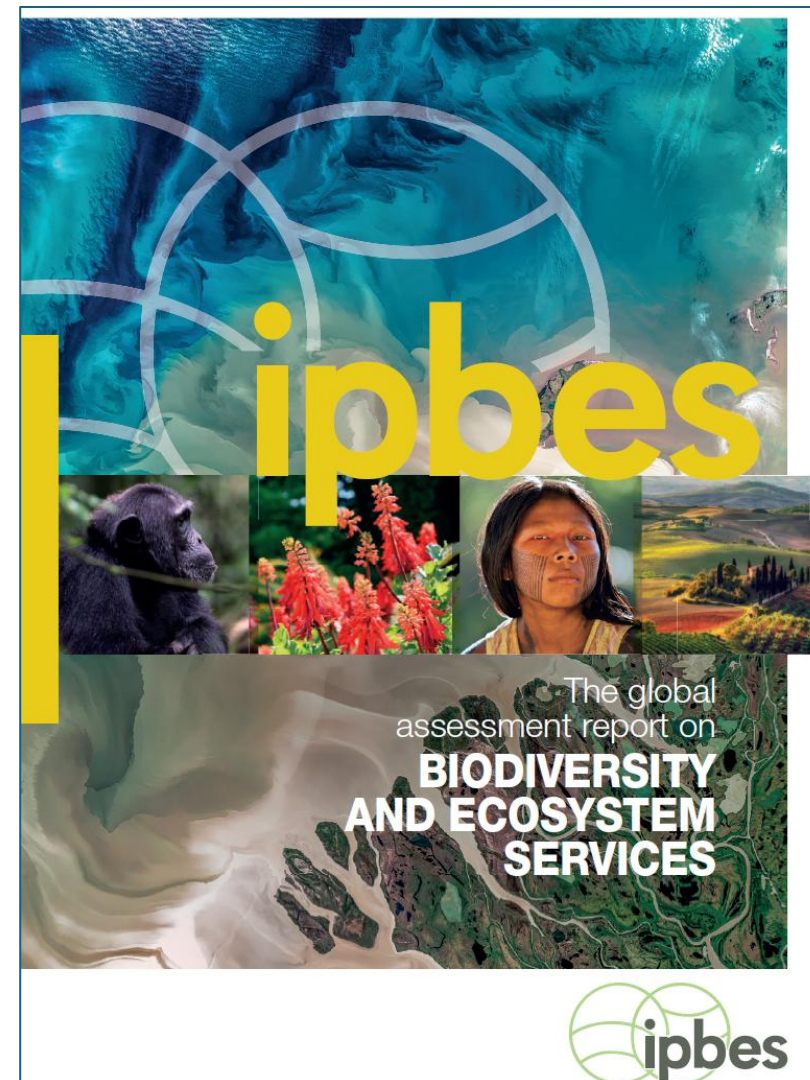
Mace, 2014, *Science*



Nature and ecosystems are fundamental to human wellbeing

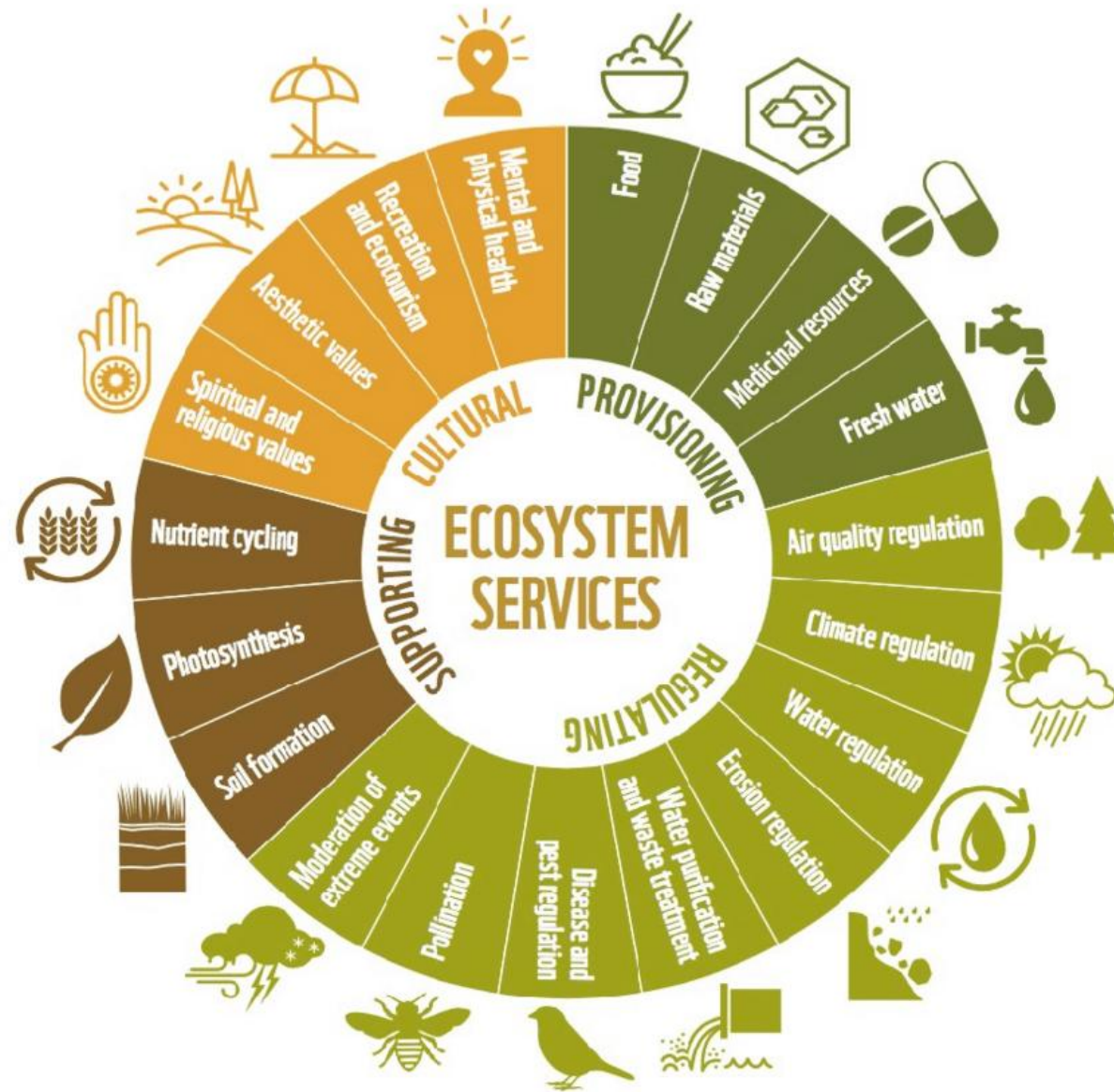


Millennium Ecosystem Assessment, 2005

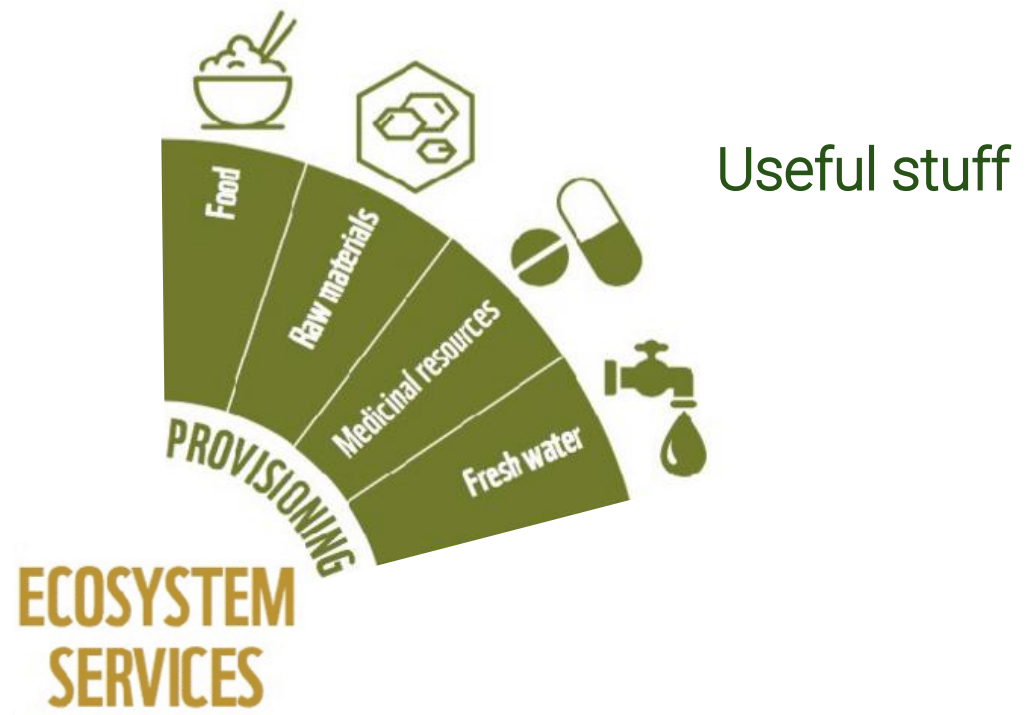


IPBES Global Assessment, 2019

Nature and ecosystems are fundamental to human wellbeing



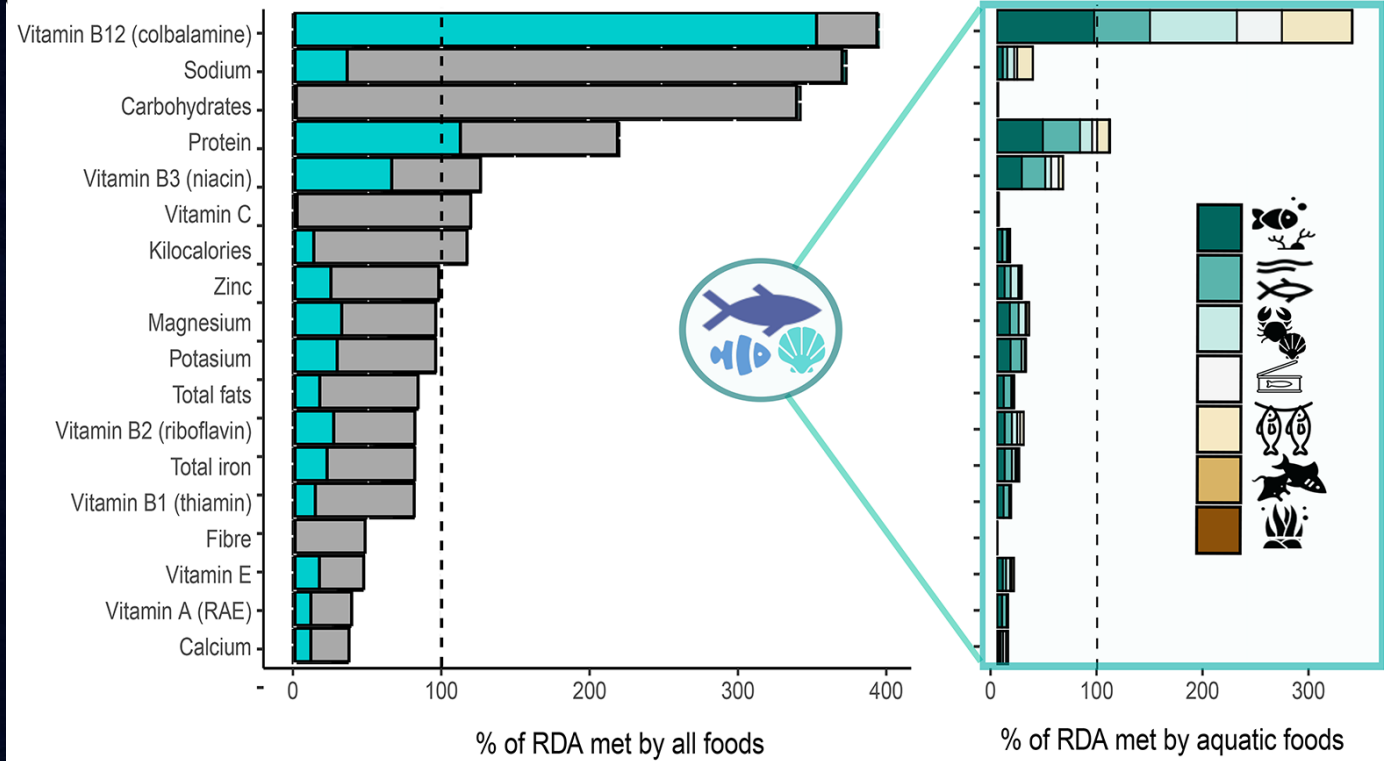
“Ecosystem services” = the benefits that people receive from ecosystems



Ecosystems support food security

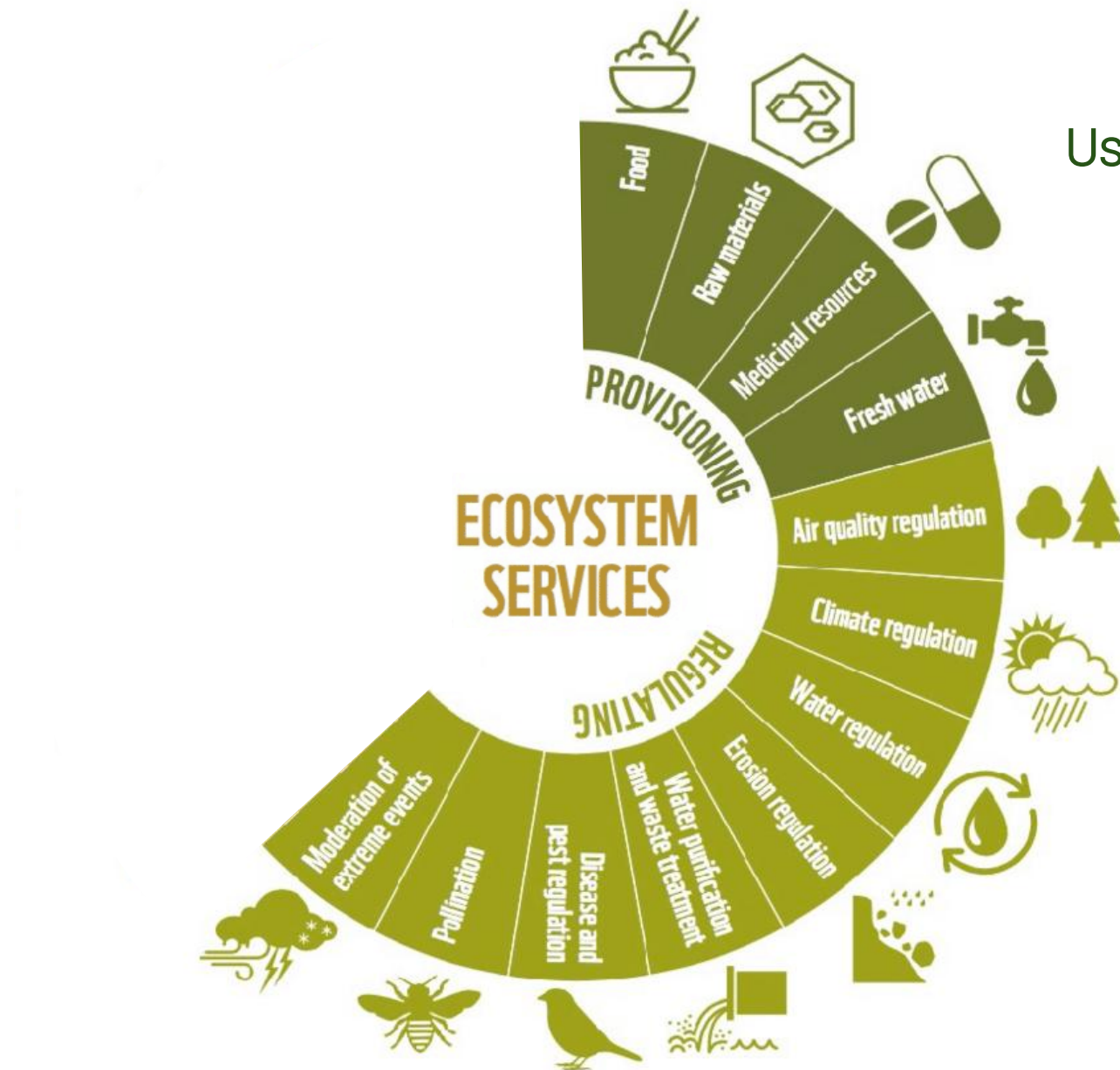


Image: European Space Agency



“Aquatic foods [from coral reefs] are a critical source of nutrients for Small Island Developing States such as the Republic of Kiribati”

Zamborain-Mason et al. 2025, Sci Rep



Useful stuff

A stable and healthy environment

More intact ecosystems suppress the risk of Lyme disease



Forest degradation
and fragmentation



More small
mammal hosts of
Lyme bacterium
(e.g. *white-footed mouse*)

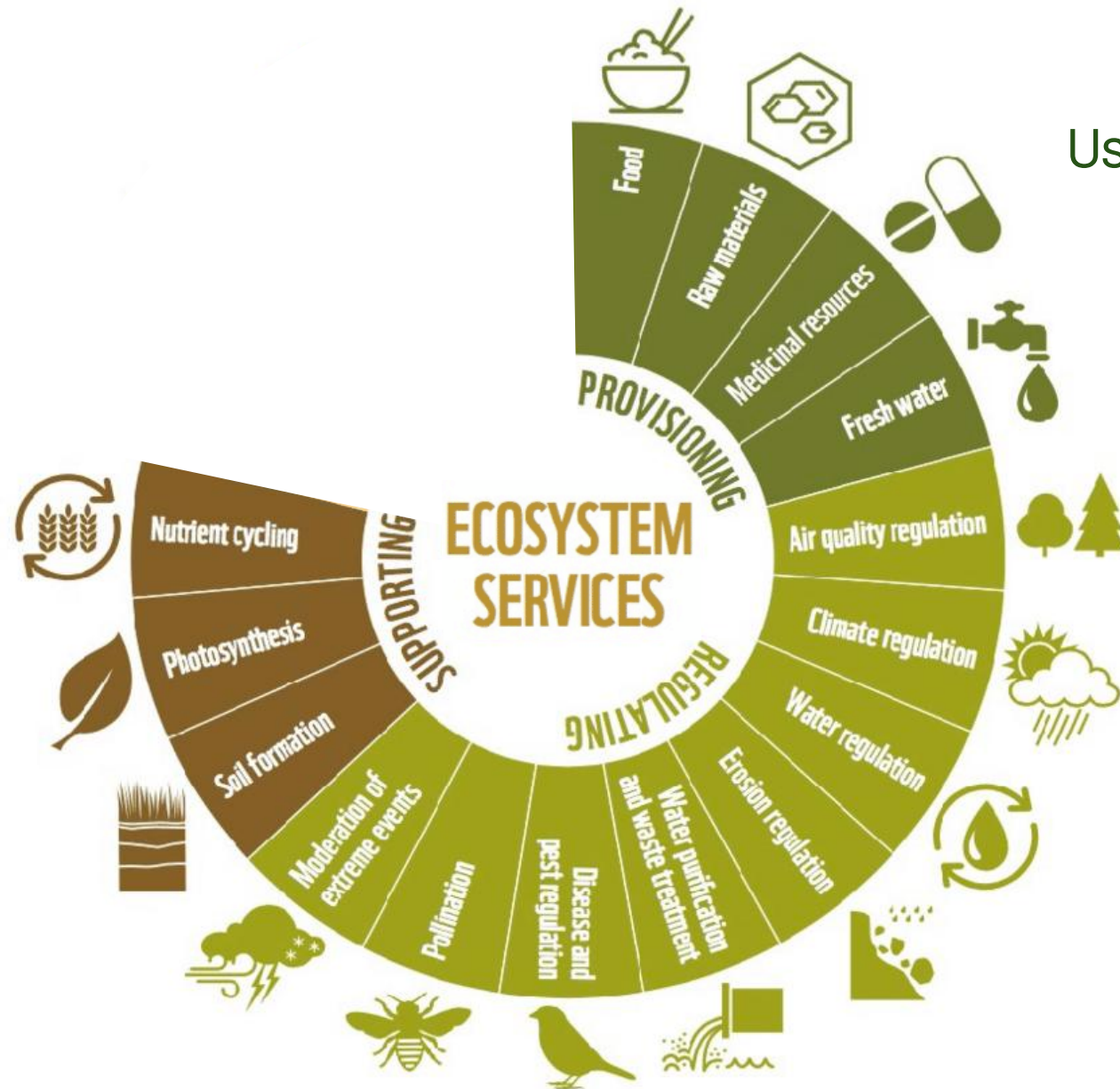


More infectious
ticks



More people
getting sick

Processes
fundamental to
life as we know it



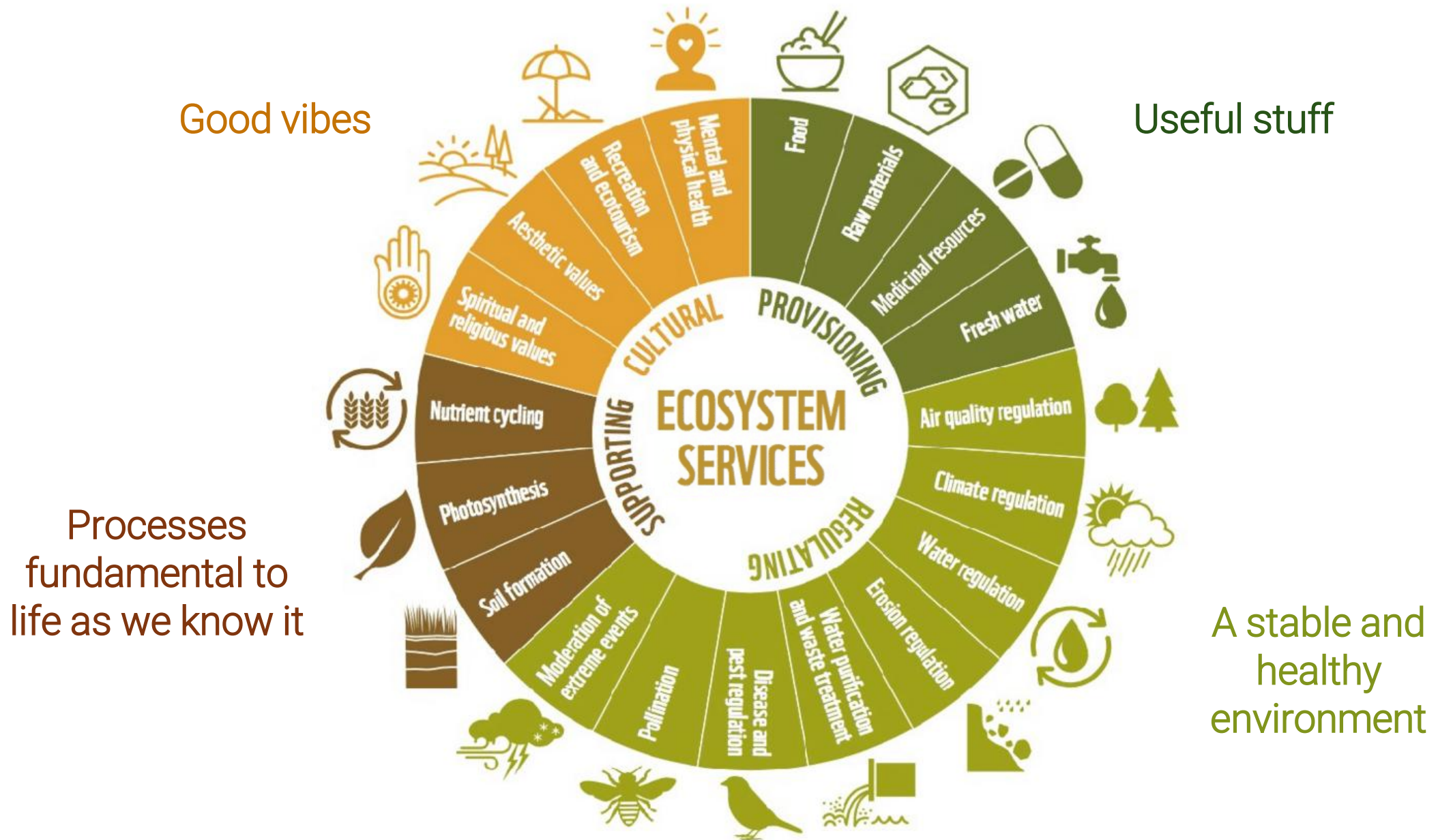
Useful stuff

A stable and
healthy
environment

Clearing and intensive use of land drives soil degradation

“Roughly 1.7 billion people are living in areas where crop yields are failing due to human-induced land degradation – a pervasive and silent crisis that is undermining agricultural productivity and threatening ecosystem health worldwide”

UN Food and Agriculture Organization (FAO)





A woman with long dark hair tied back, wearing a light-colored jacket, stands in a forest. She is looking upwards with her head tilted back, her arms slightly raised. The background is filled with the dense canopy of trees, with sunlight filtering through the leaves, creating a soft, dappled light effect. The overall mood is peaceful and contemplative.

“more ambient greenness around the home [was] associated with a reduced likelihood of having a common mental health disorder”

THE LANCET
Planetary Health

Ambient greenness, access to local green spaces, and subsequent mental health: a 10-year longitudinal dynamic panel study of 2·3 million adults in Wales

[Rebecca S Geary, PhD^a](#) · [Daniel Thompson, PhD^b](#) · [Amy Mizen, PhD^b](#) · [Ashley Akbari, MSc^b](#) · [Joanne K Garrett, PhD^d](#) ·

What happens as ecosystems are degraded?

What happens as ecosystems are degraded?



Lough Neagh, Northern Ireland

- Increasingly **homogeneous** and dominated by a few species
- Animal and plant populations live with **multiple physiological stresses**
- **Less productive and lose key functions** (*including those key to humans e.g. pollination*)
- Prone to **abrupt collapse** or changes in state (*e.g. from healthy to algal-dominated*)

What could global ecological degradation mean for human health?



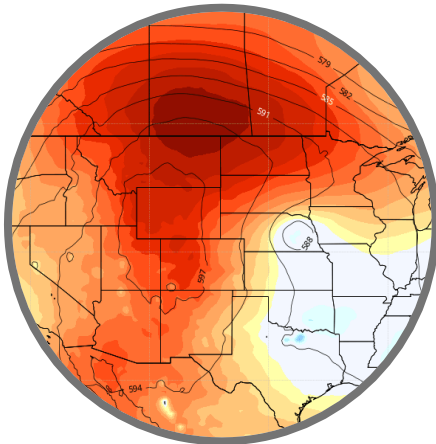
**Loss of clean water
and food provision**



**Vulnerability to food
system shocks**



**New and resurgent
epidemics**



**Exposure to
climate extremes**

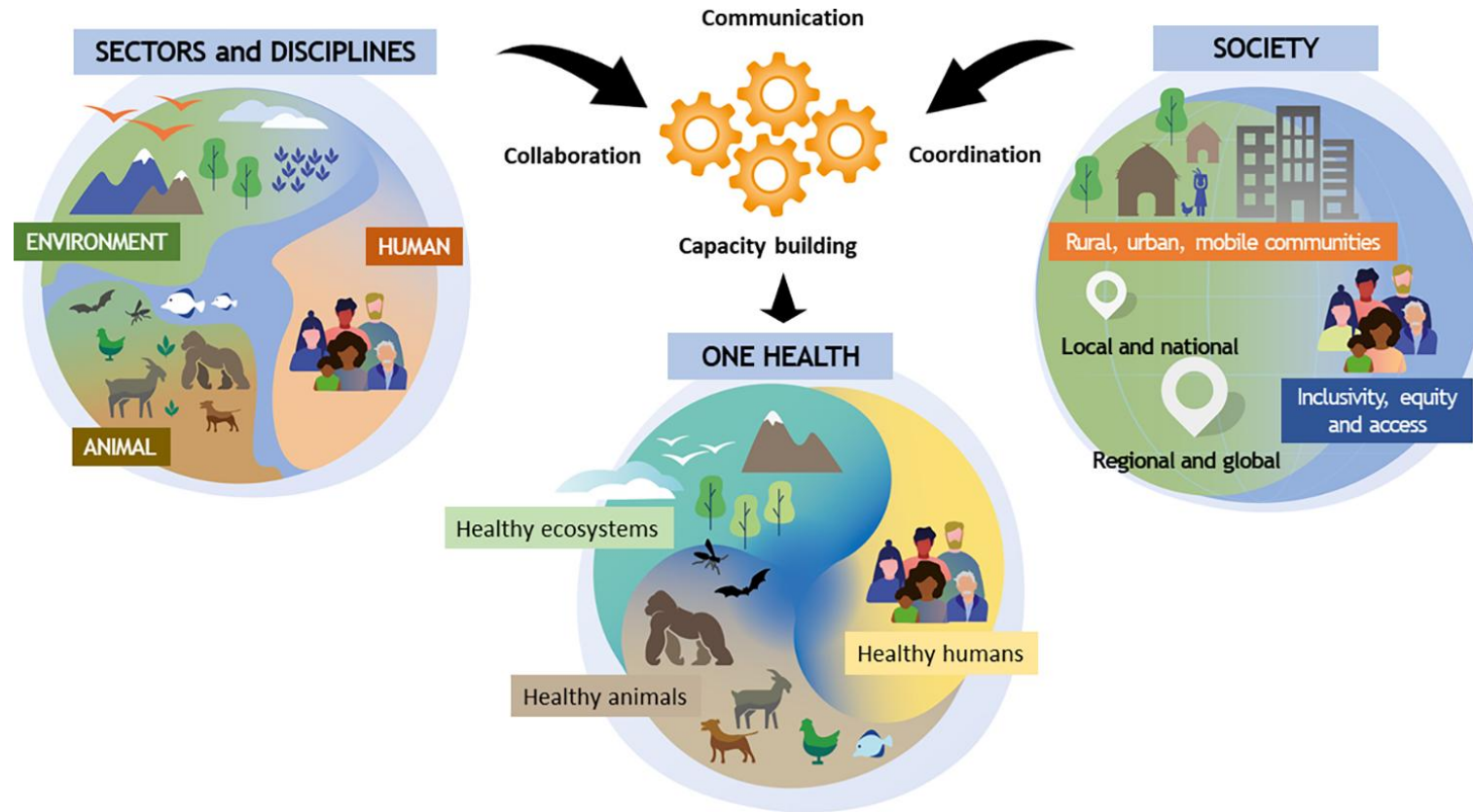


**Impacts on mental
health**



**Greater risks of
disasters**

The “One Health” concept: human, animal and ecosystem health are inseparable

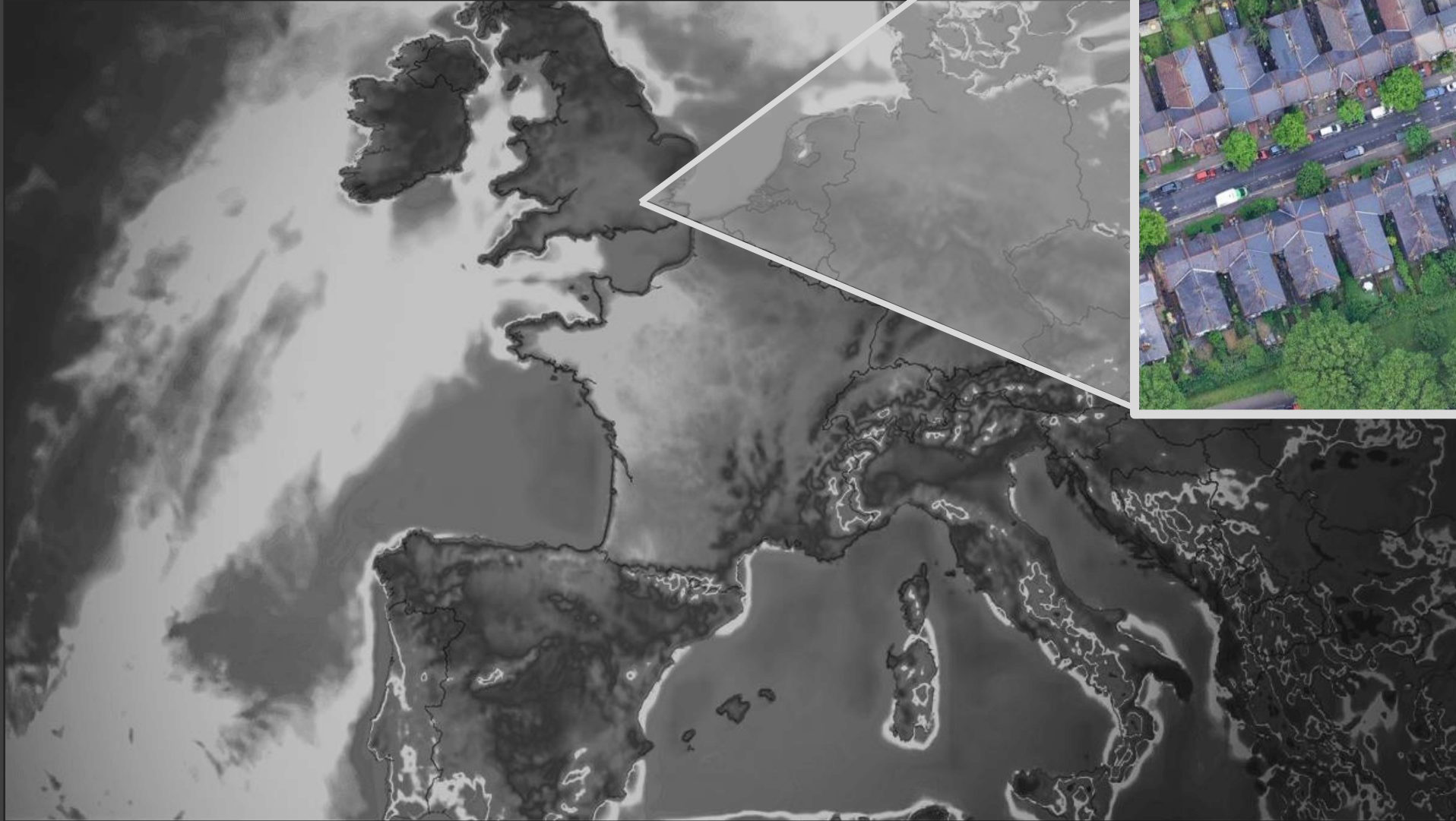


“One Health is an integrated, unifying approach that aims to sustainably balance and optimize the health of people, animals, and ecosystems. It recognizes the health of humans, domestic and wild animals, plants, and the wider environment (including ecosystems) are closely linked and interdependent.”

(One Health High-Level Expert Panel, 2022, PLoS Pathogens)

Putting these ideas into practice:
mosquito-borne diseases

Dusk, somewhere in East London,
during a heatwave...



16 (attempted) bites in 10 minutes!



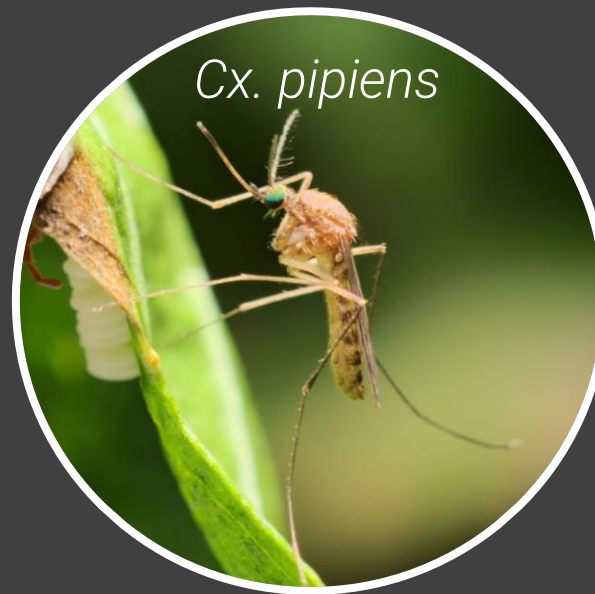


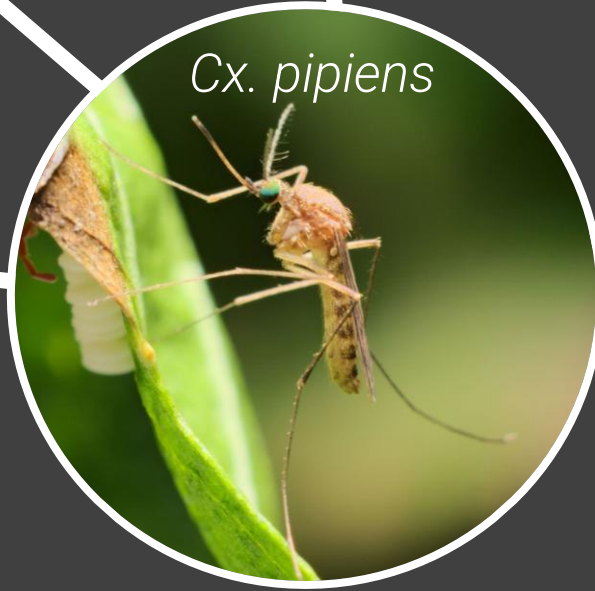
House mosquito
(*Culex pipiens*)

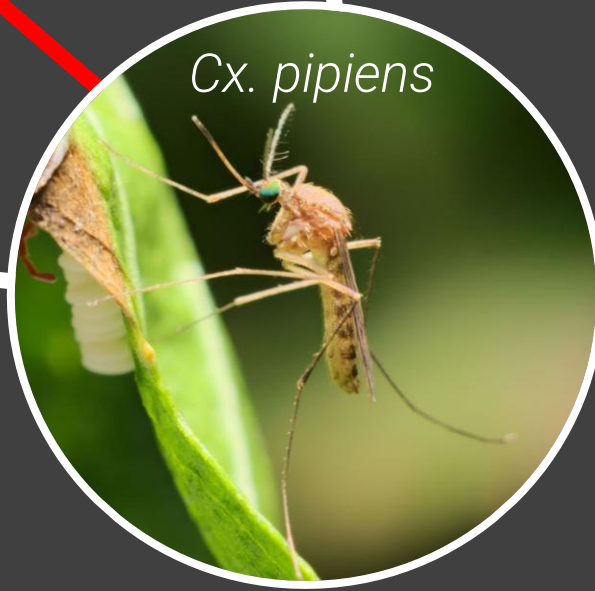




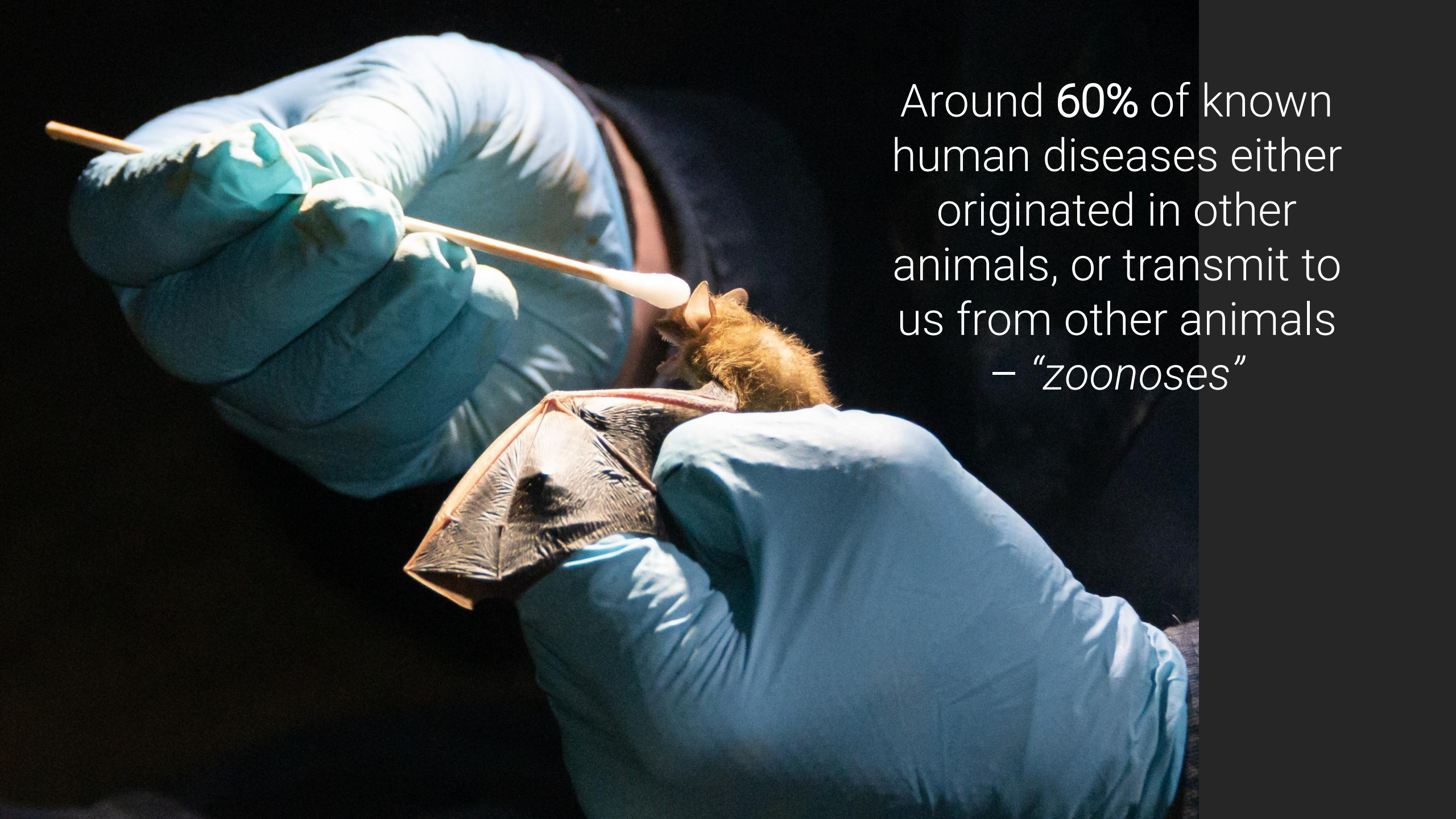
Why should we pay attention to mosquitoes?



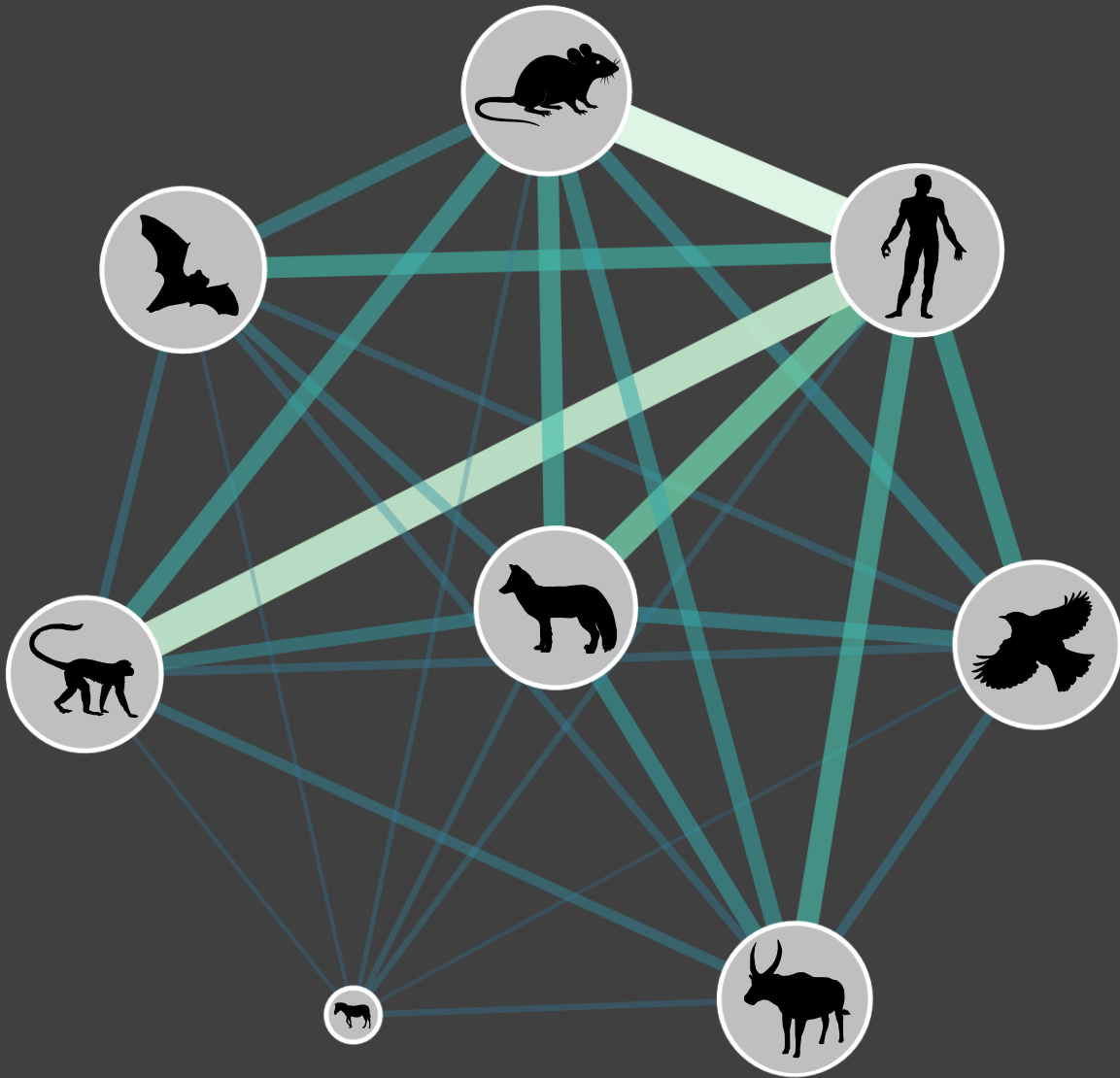






A close-up photograph showing a person wearing blue nitrile gloves. One hand holds a small bat by its wings and body, while the other hand uses a white cotton swab to collect a sample from the bat's mouth. The bat has brown fur and large, dark wings. The background is dark, and the scene is illuminated by a focused light source.

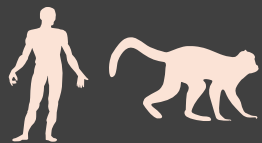
Around 60% of known
human diseases either
originated in other
animals, or transmit to
us from other animals
– “zoonoses”



Around 60% of known human diseases either originated in other animals, or transmit to us from other animals – “zoonoses”

Known patterns of virus sharing between animal groups: thicker links = more viruses in common

(Data: Carlson, Gibb et al. 2021, mBio)



Dengue fever



Yellow fever



Japanese encephalitis



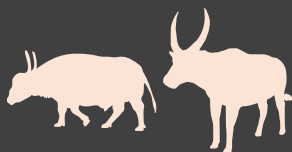
Malaria



Zika fever



West Nile fever



Rift Valley fever

Mosquitoes transmit
around 80 known
human diseases –
most of which come
from other animals





Dengue fever



Yellow fever



Japanese encephalitis



Malaria



Zika fever



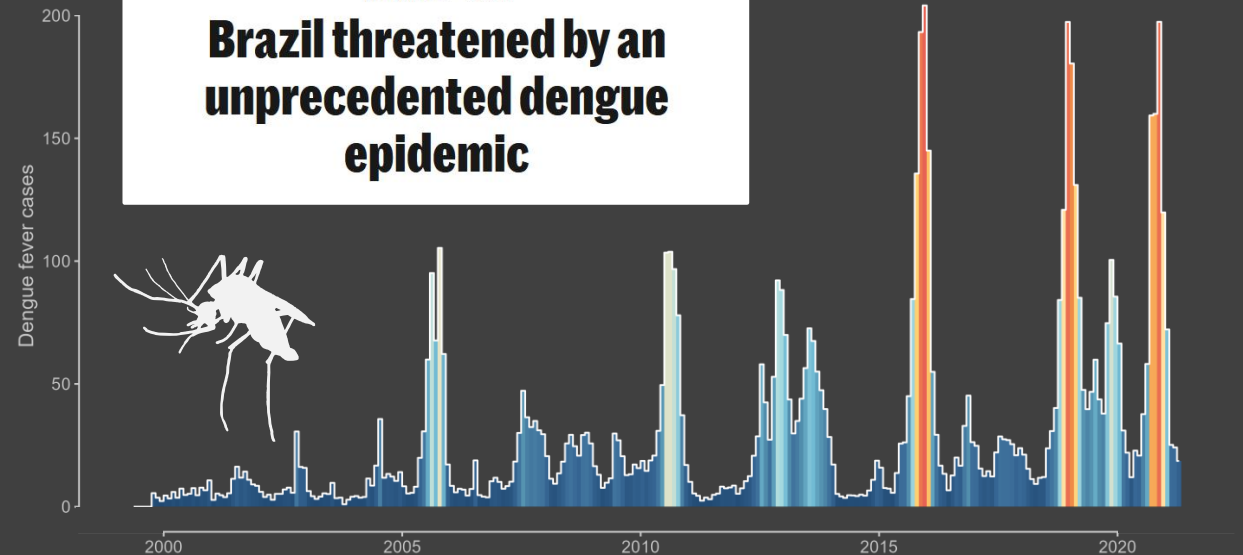
West Nile fever



Rift Valley fever



14.4 million
dengue cases
reported
globally in
2024

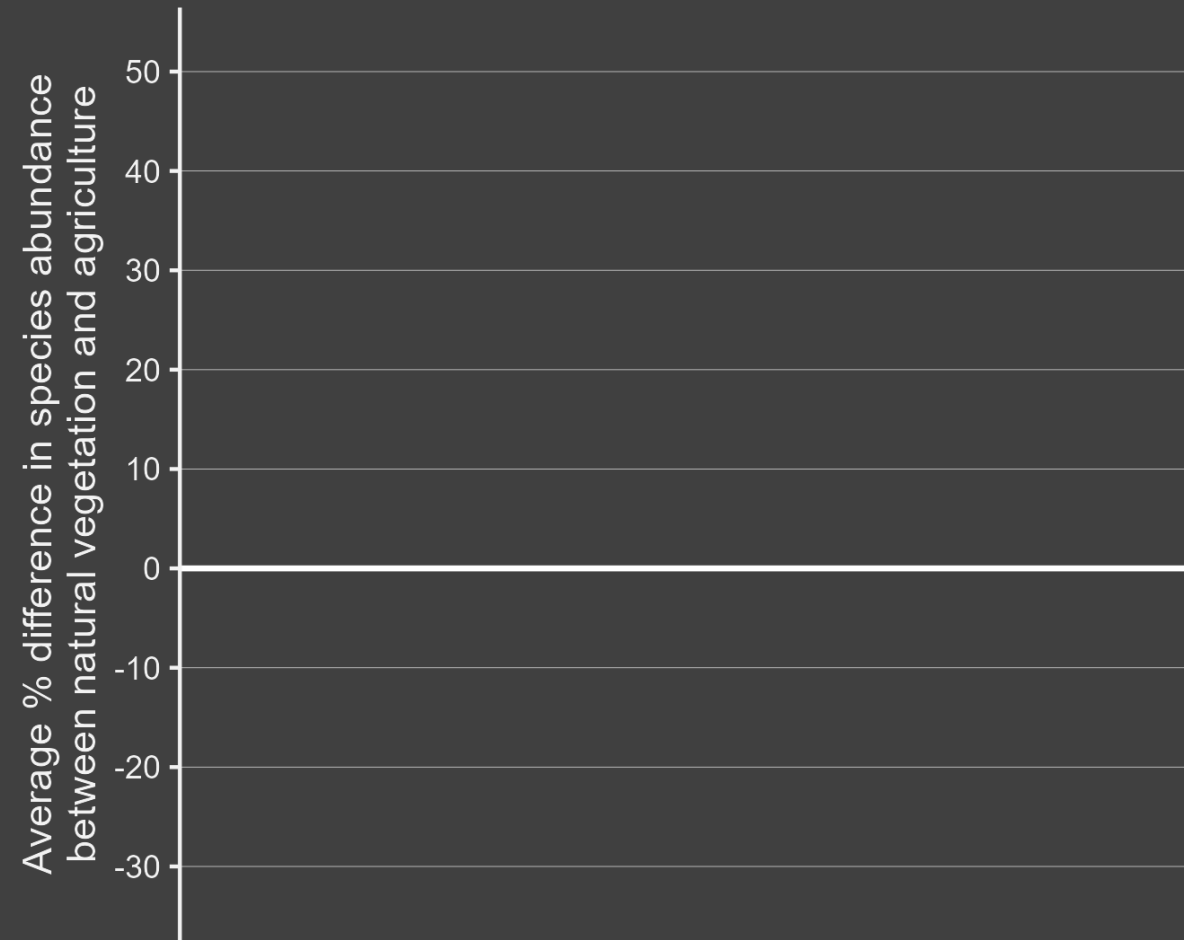


Mosquitoes (and their pathogens)
remind us that humans are not
separate from “nature” – we are
embedded in the web of life



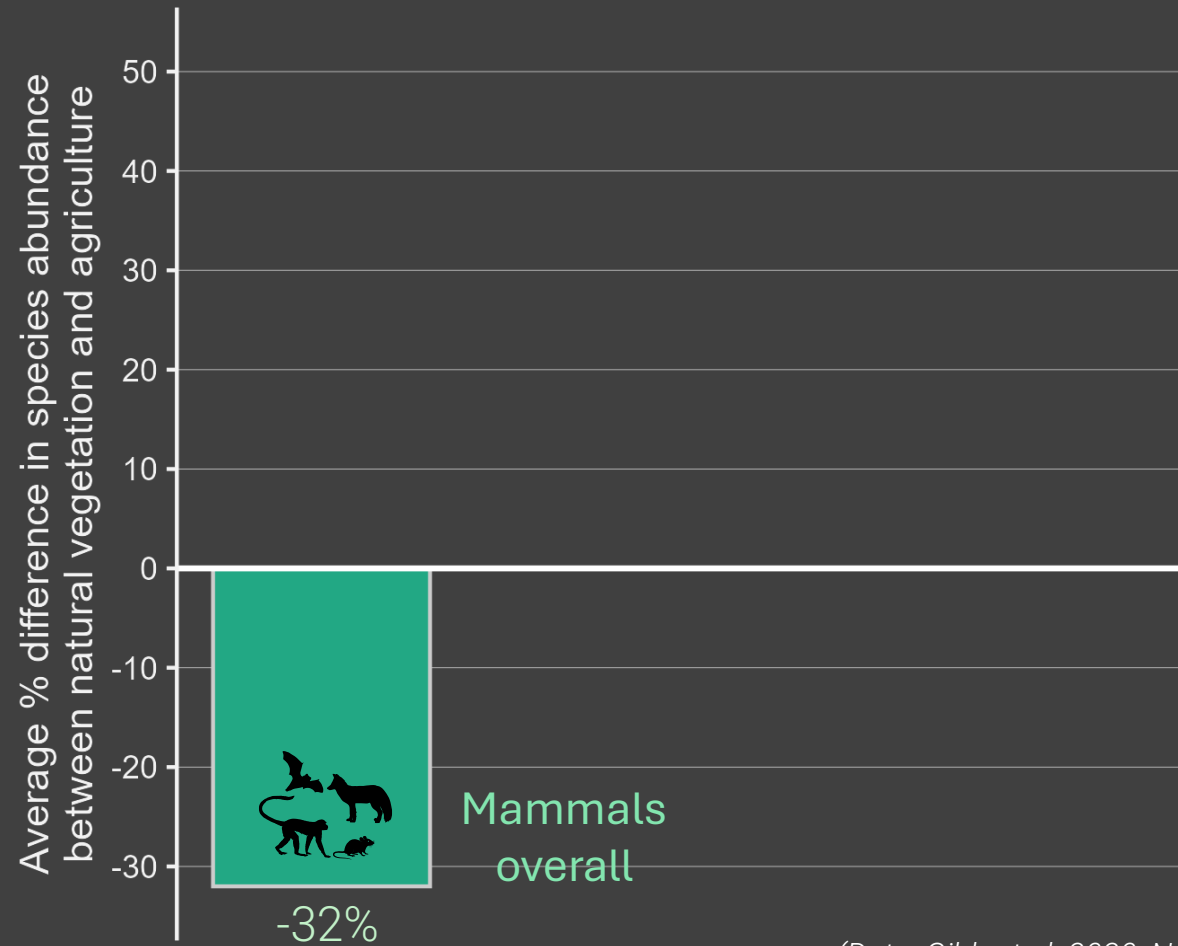
Image: Miguel Pinheiro/CIFOR

How do wildlife populations change after natural vegetation has been converted to agriculture?



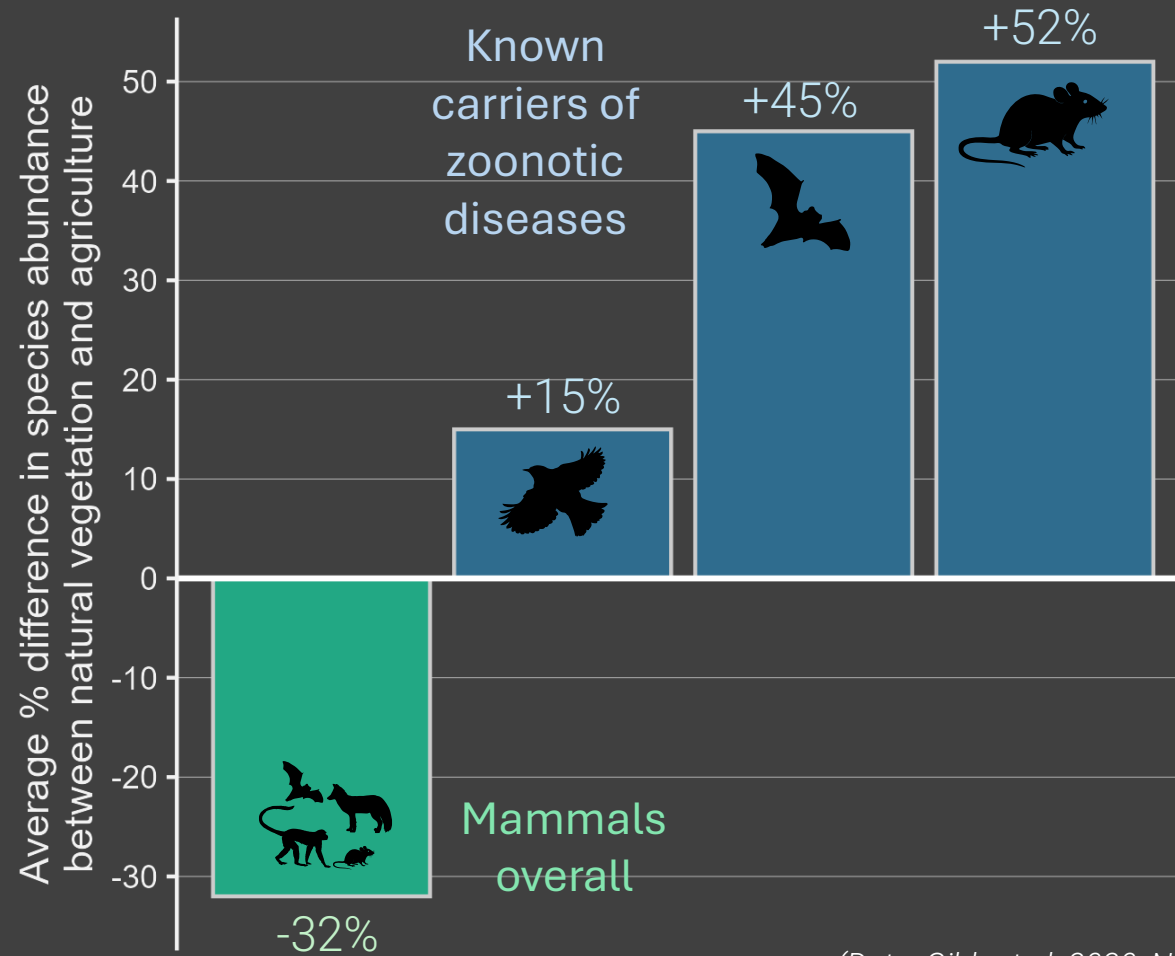
(Data: Gibb et al. 2020, Nature)

How do wildlife populations change after natural vegetation has been converted to agriculture?



(Data: Gibb et al. 2020, Nature)

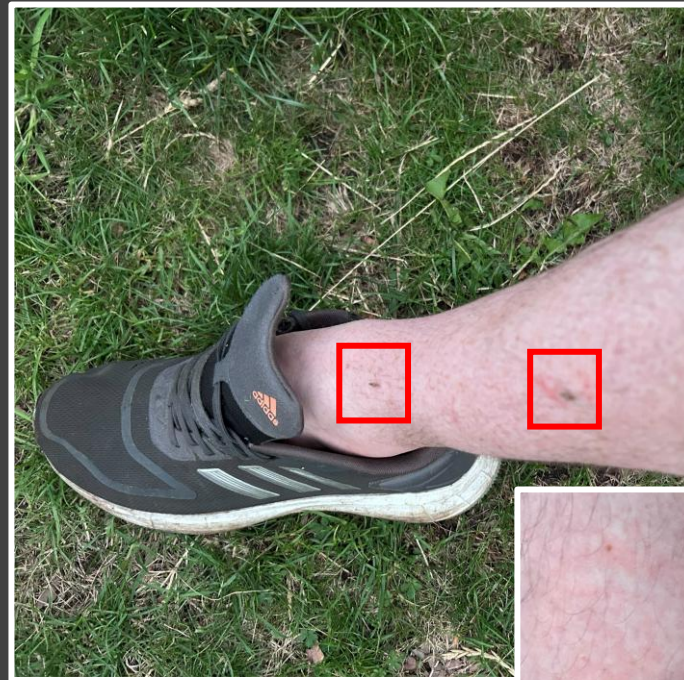
How do wildlife populations change after natural vegetation has been converted to agriculture?

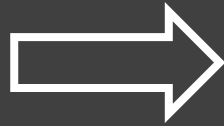
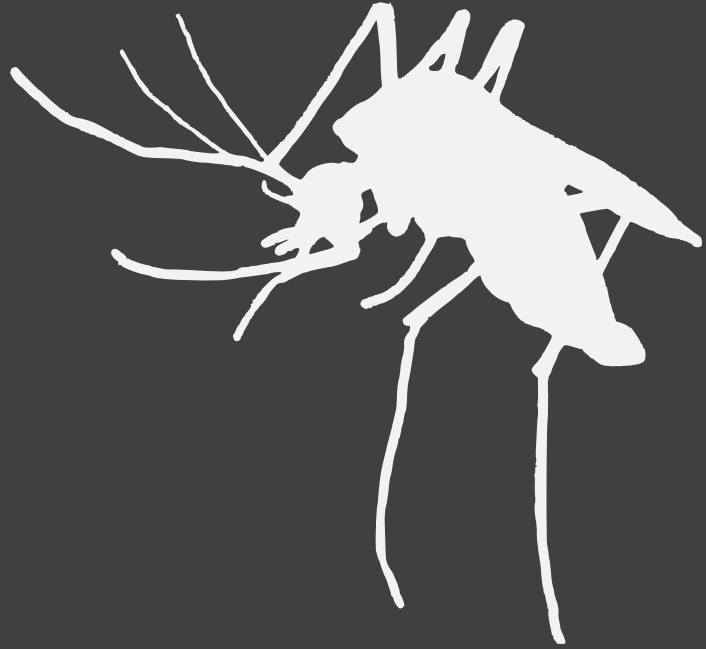


(Data: Gibb et al. 2020, Nature)

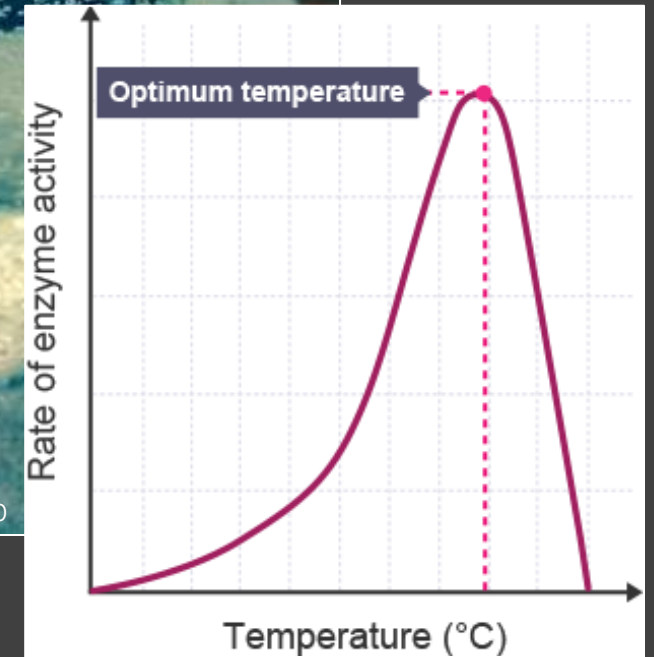
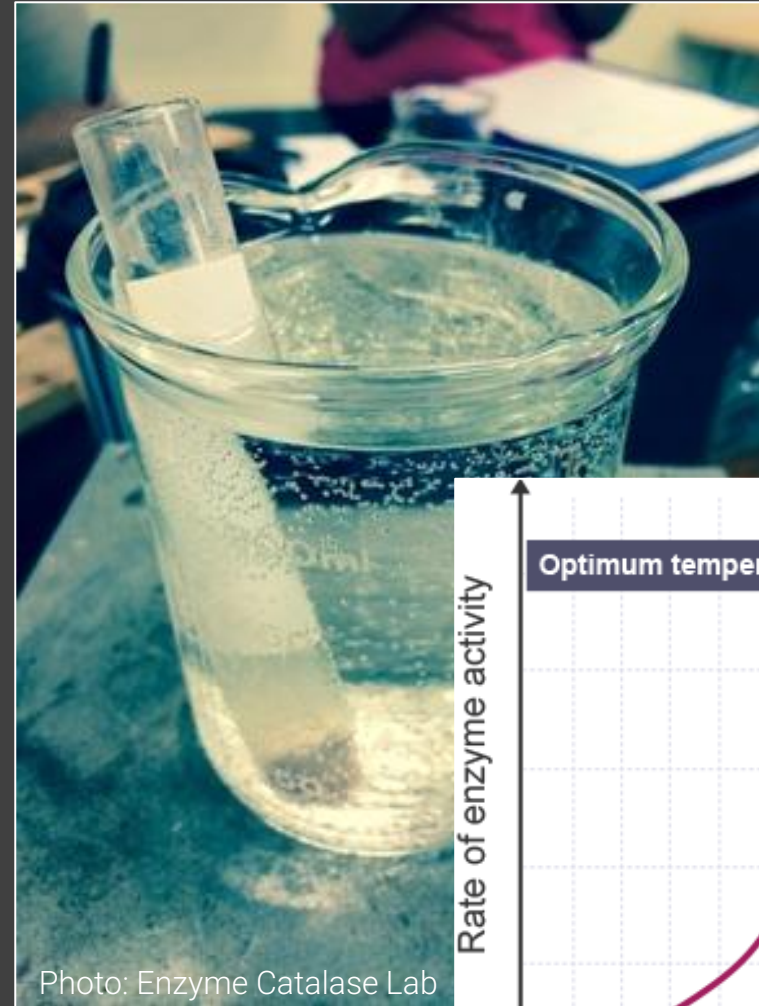


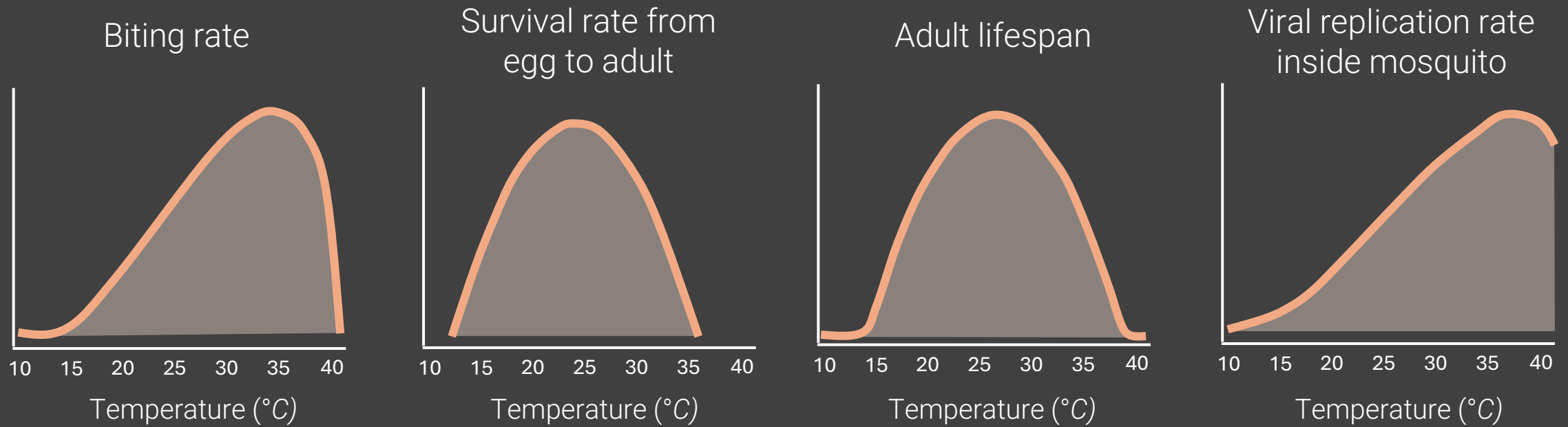
Heatwave experiment, 24-26 June:
average of 14 bites in 10 minutes
(nighttime temperatures $\sim 19\text{-}22^{\circ}\text{C}$)





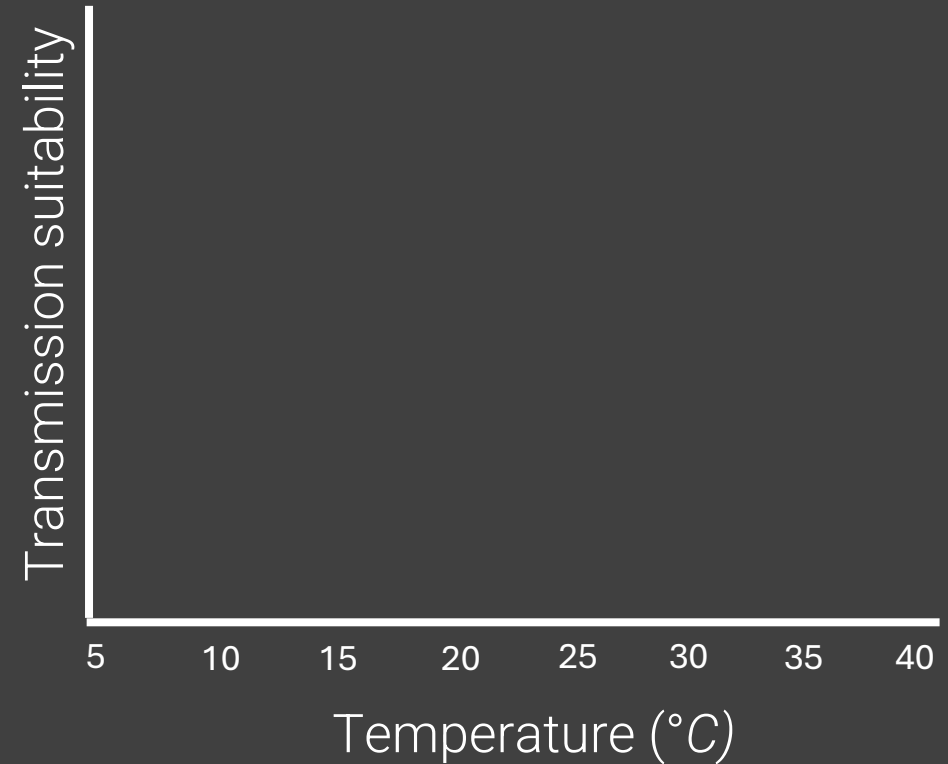
Mosquitoes are ectotherms:
their internal chemistry occurs at
the same temperature as the
outside environment



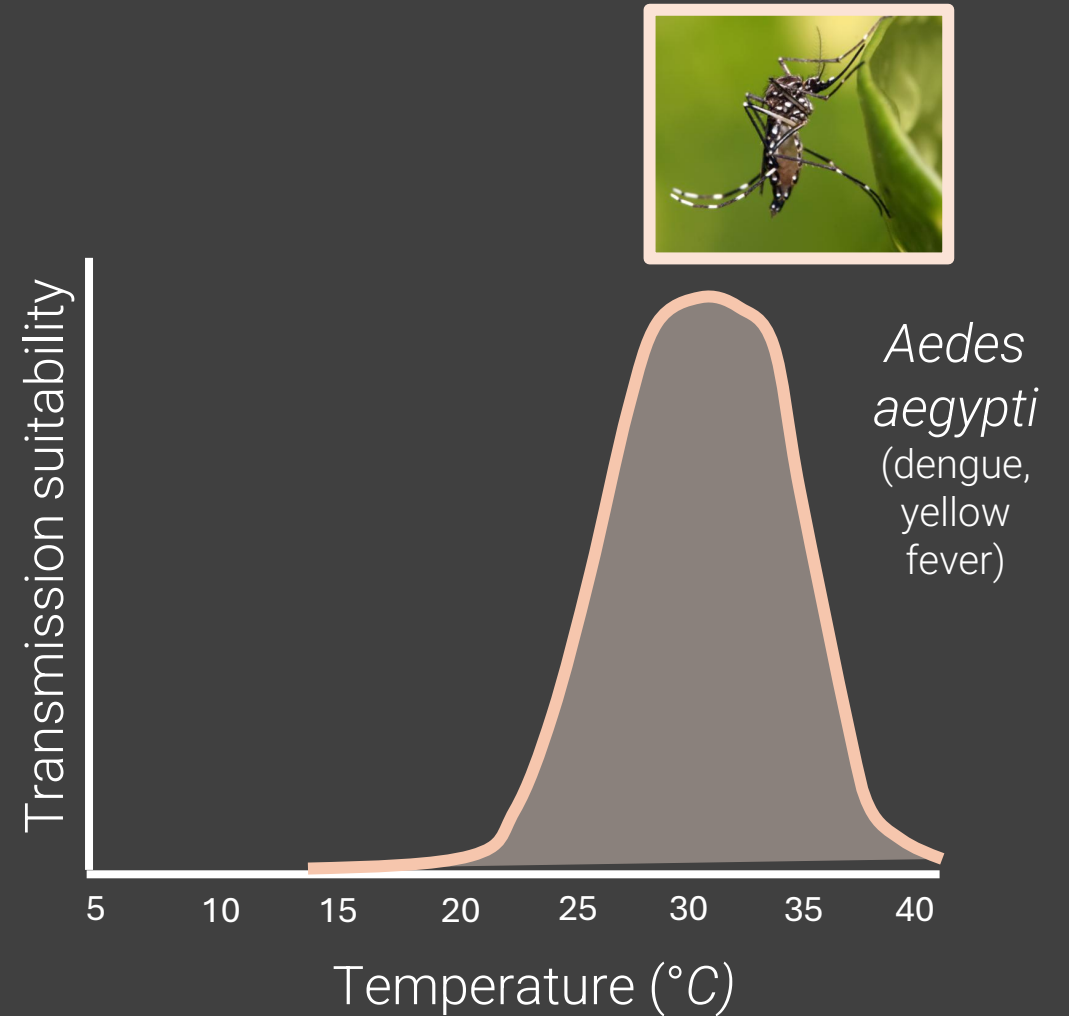


(For *Aedes aegypti* and dengue virus, adapted from Mordecai et al. 2017)

Mosquitoes are most abundant and effective at transmitting pathogens within an optimum temperature range

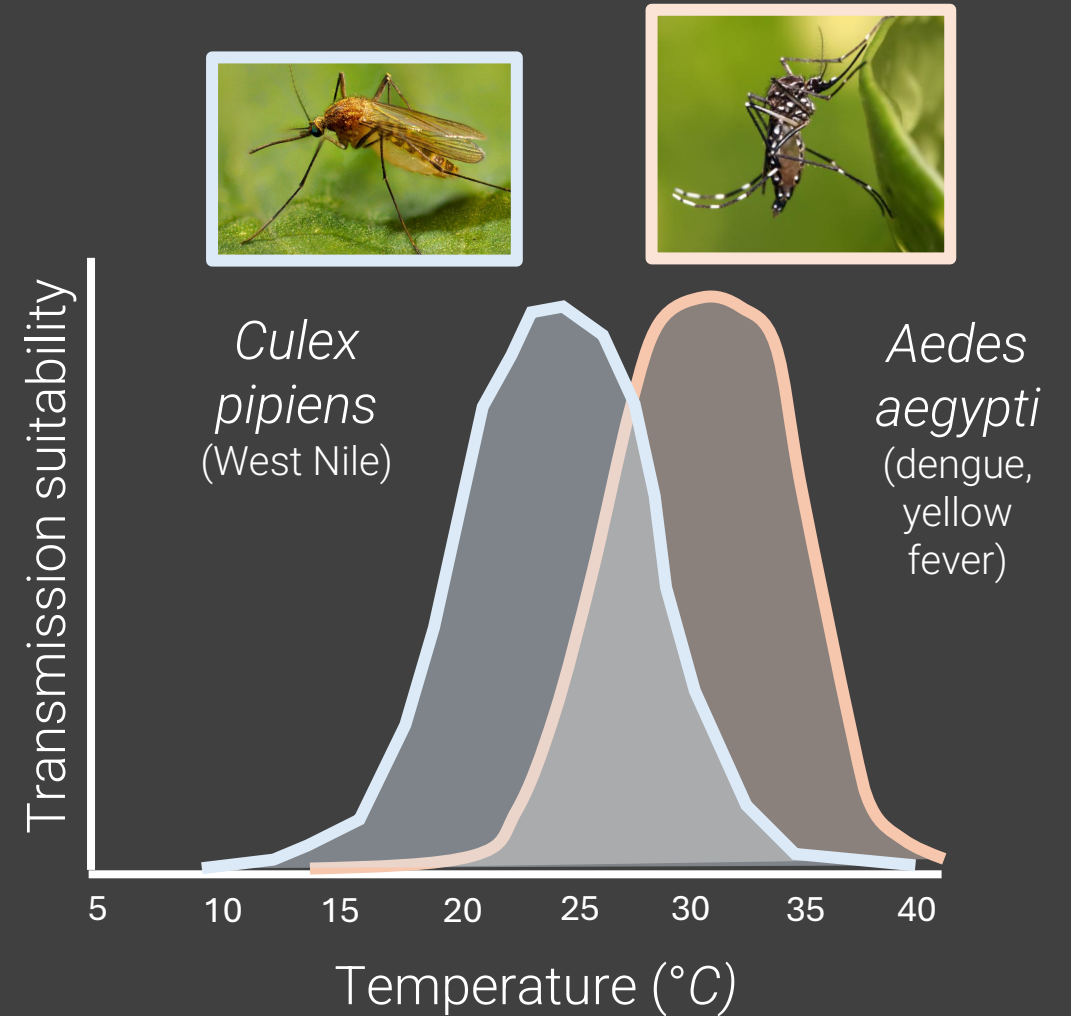


Mosquitoes are most abundant and effective at transmitting pathogens within an optimum temperature range

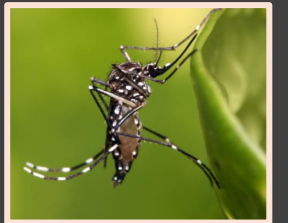
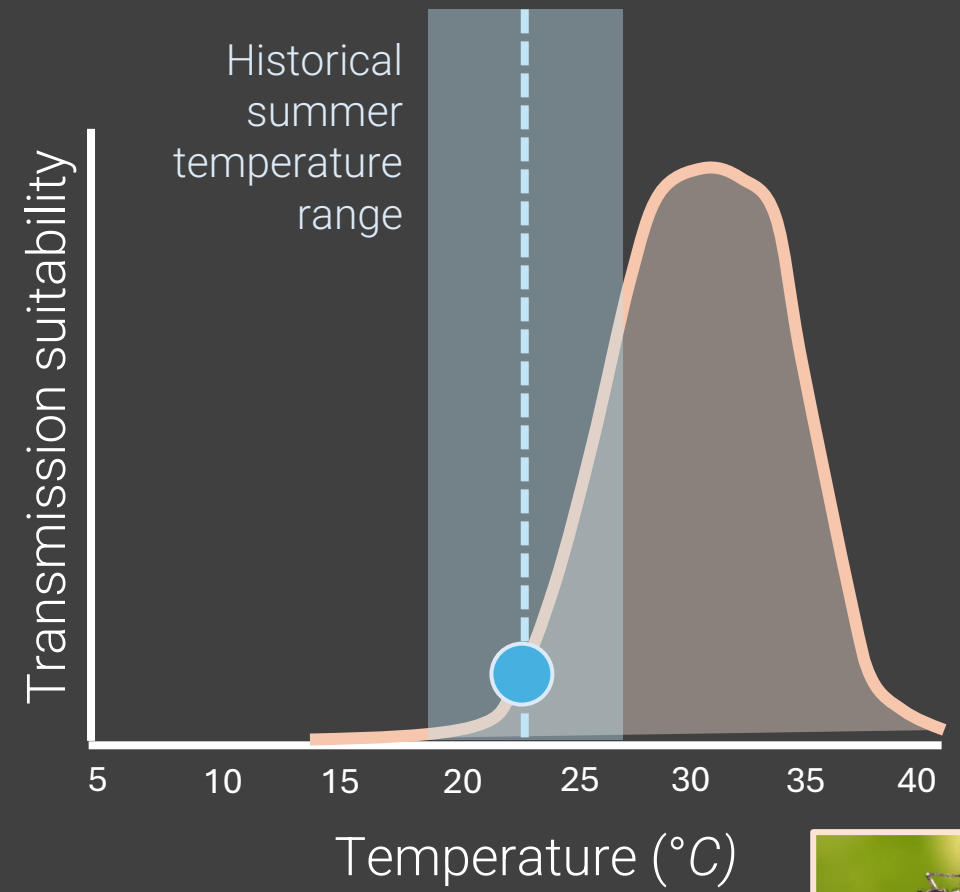


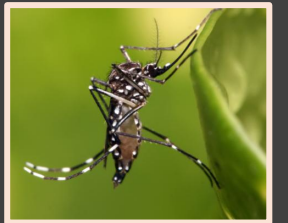
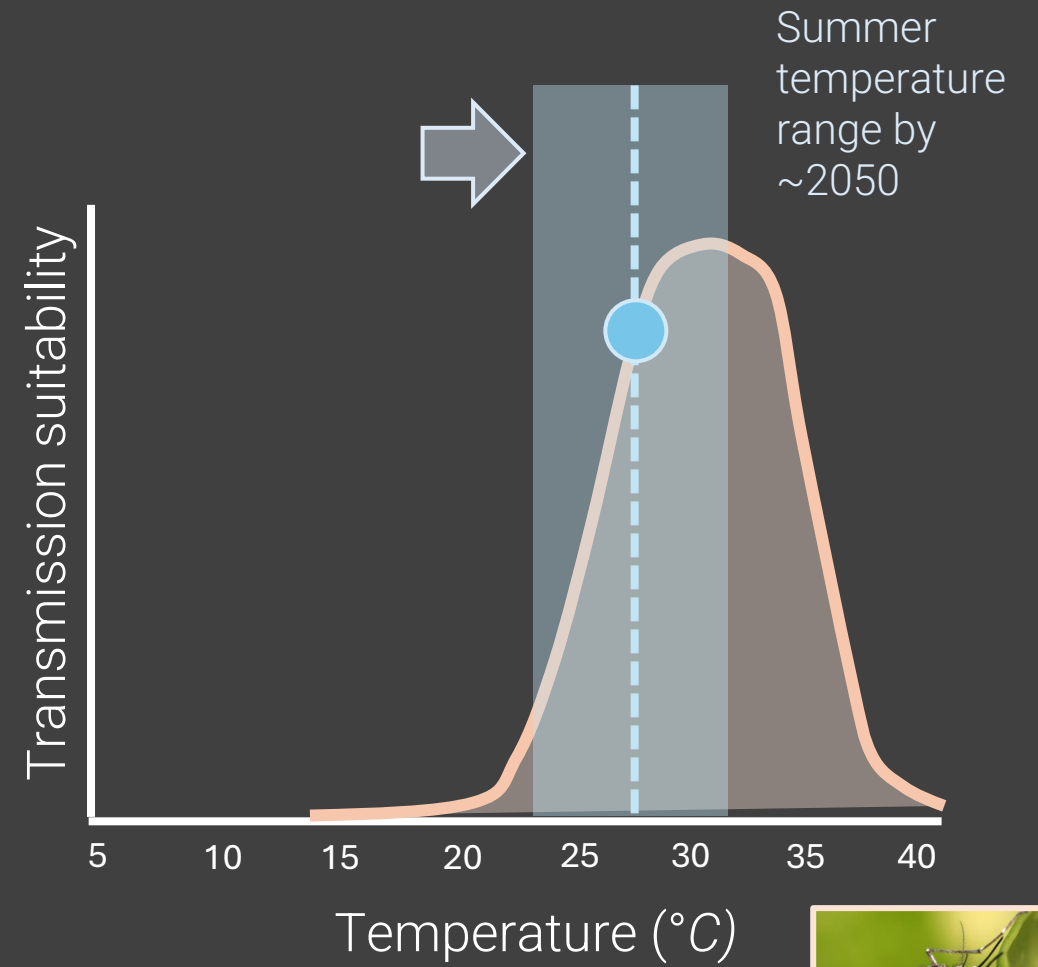
(Adapted from Mordecai et al. 2019)

Mosquitoes are most abundant and effective at transmitting pathogens within an optimum temperature range

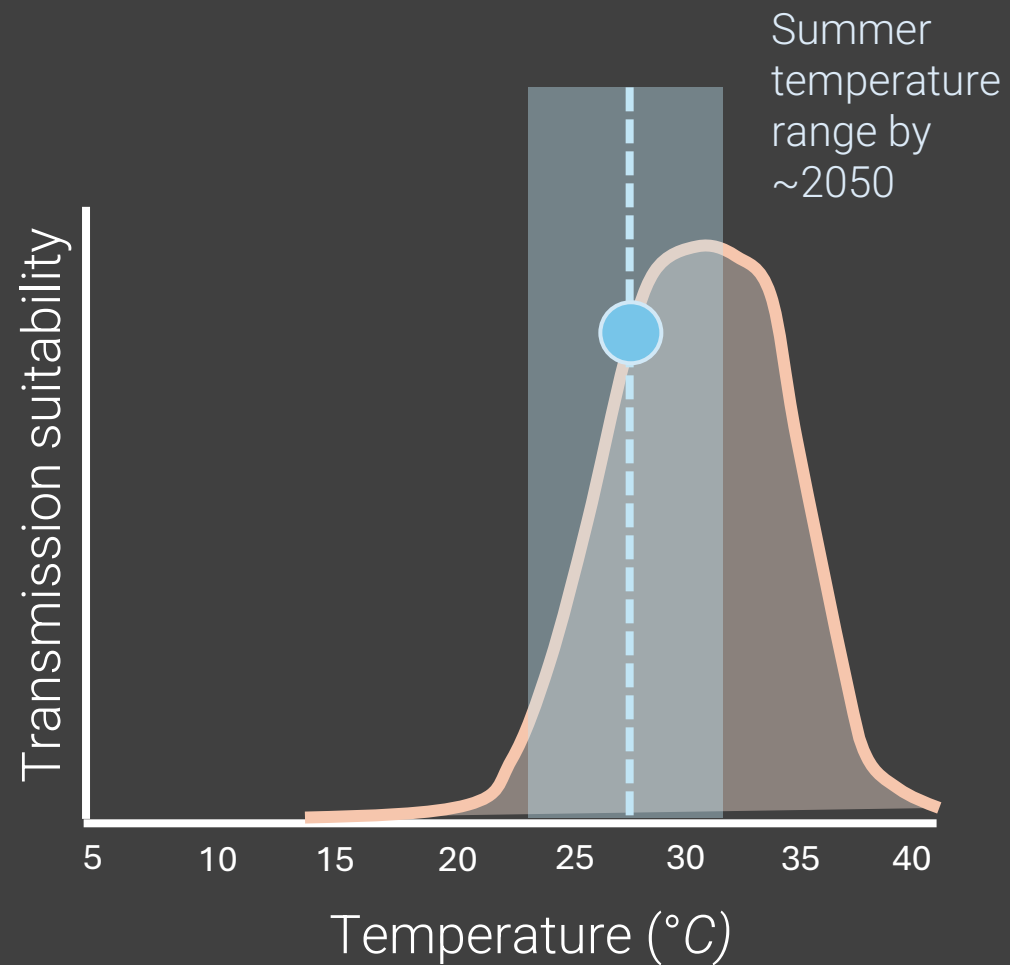
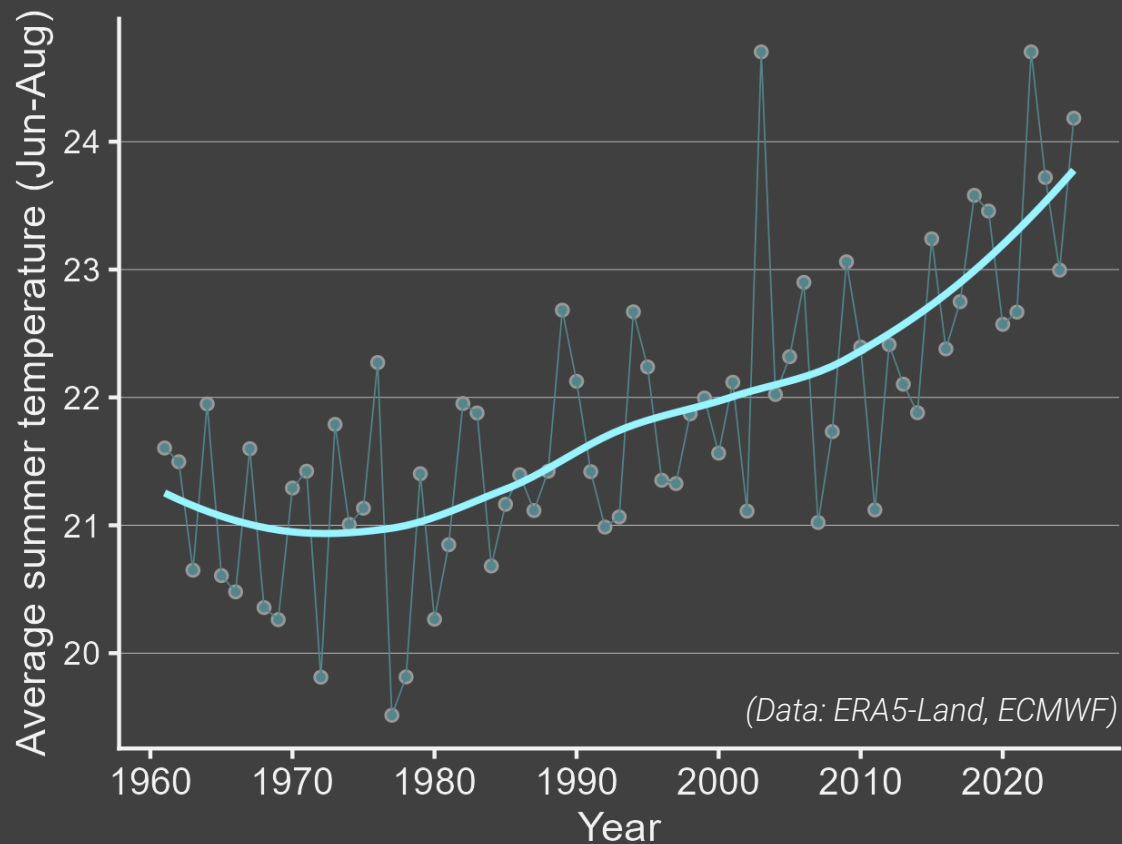


(Adapted from Mordecai et al. 2019)





Trend in summer temperatures 1960-2025,
Occitanie, southern France



GLOBAL HEALTH

A Debilitating Virus Surges Globally as Mosquitoes Move With Warming Climate

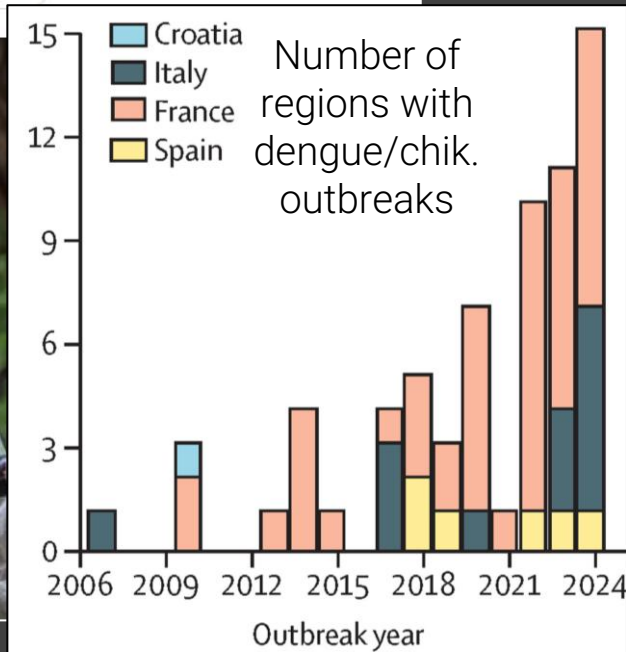
Chikungunya, which can disable victims for years, is spreading rapidly, including in China, France and other places that have not seen major outbreaks before.

Listen · 8:15 min

Share full article

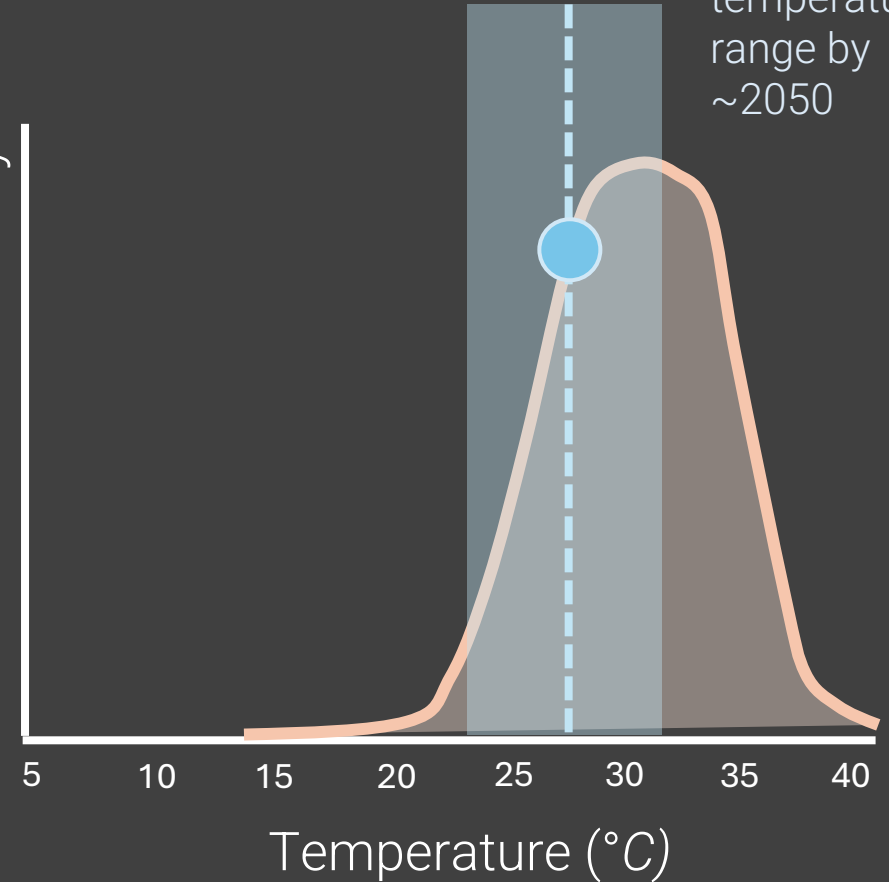


75



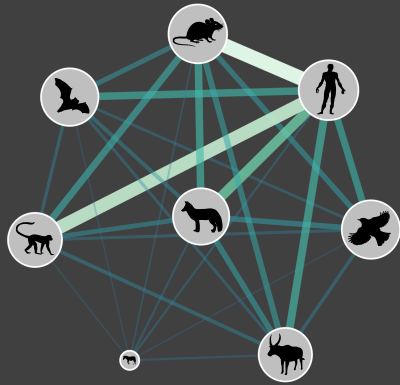
(Farooq et al. 2025)

Transmission suitability

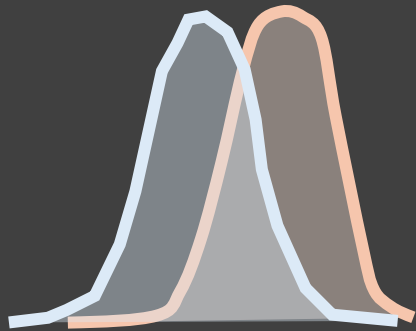


Recap: our biological understanding so far

Recap: our biological understanding so far



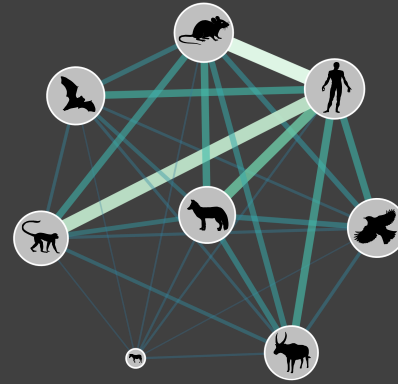
Degradation of habitats and biodiversity changes how diseases can spread between animals and humans, including via mosquitoes



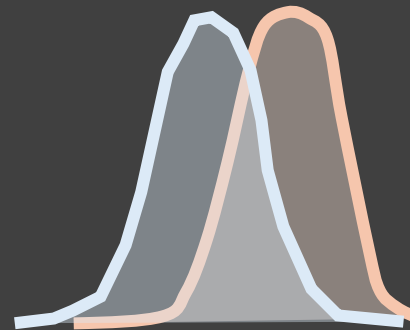
Changes in climate can directly affect how efficiently infectious diseases can spread, including through vector species such as mosquitoes

Can we measure
how much climate
change and
biodiversity loss are
affecting infectious
diseases?

Can we measure
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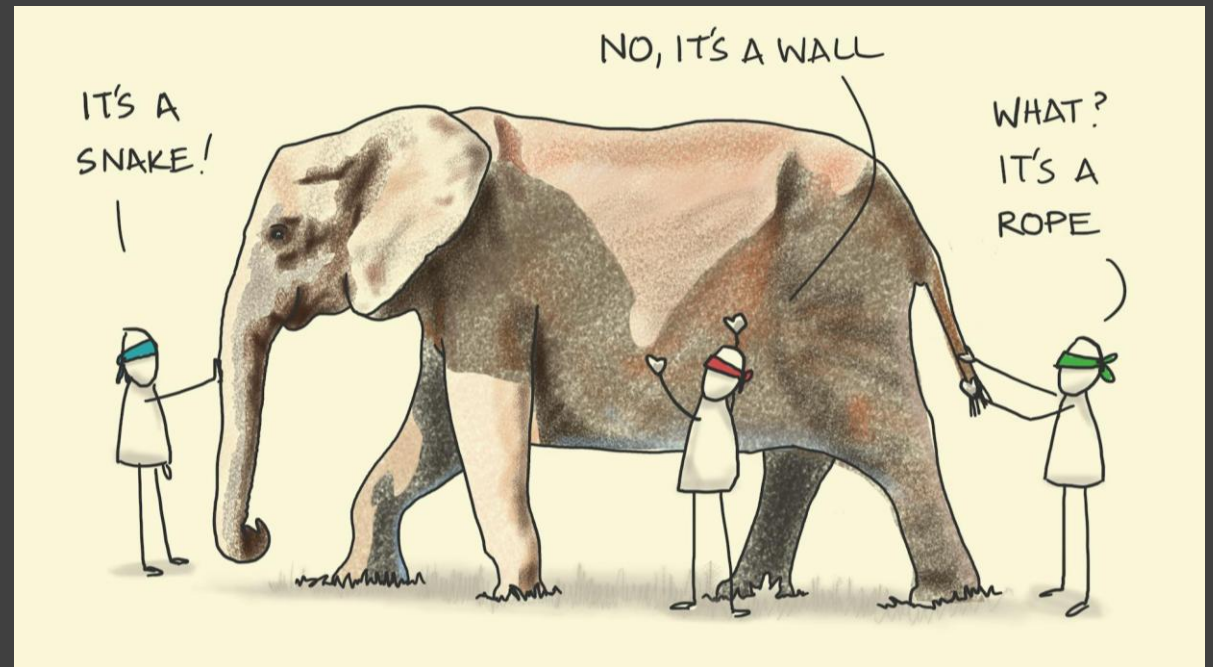


Are biodiversity loss
and land degradation
increasing the risk of
mosquito-borne
disease outbreaks?



How important is
climate warming as a
driver of mosquito-
borne diseases?

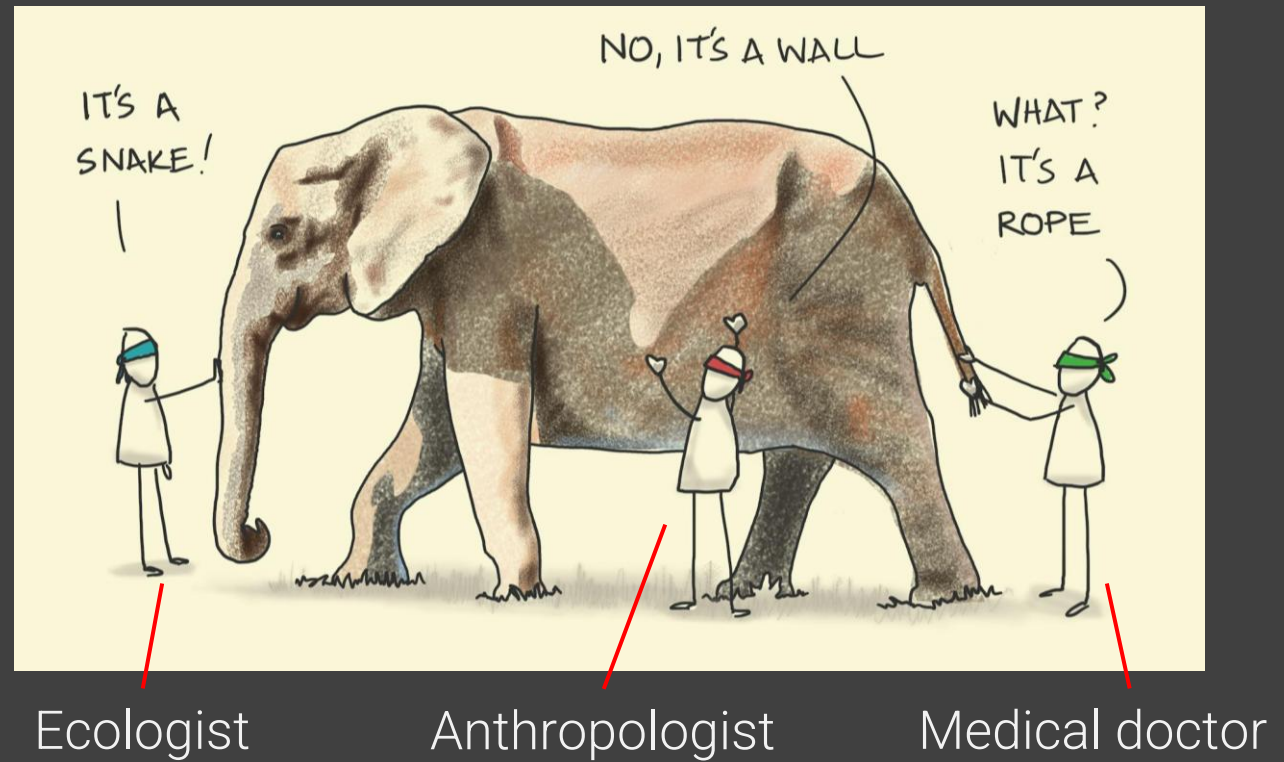
Complex problems
require
interdisciplinary
collaboration and
systems thinking



(Image: Sketchplanations)

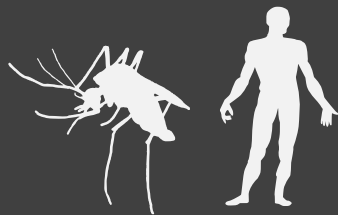
Complex problems
require
interdisciplinary
collaboration and
systems thinking

*(i.e. we need all of your diverse
ways of thinking!)*

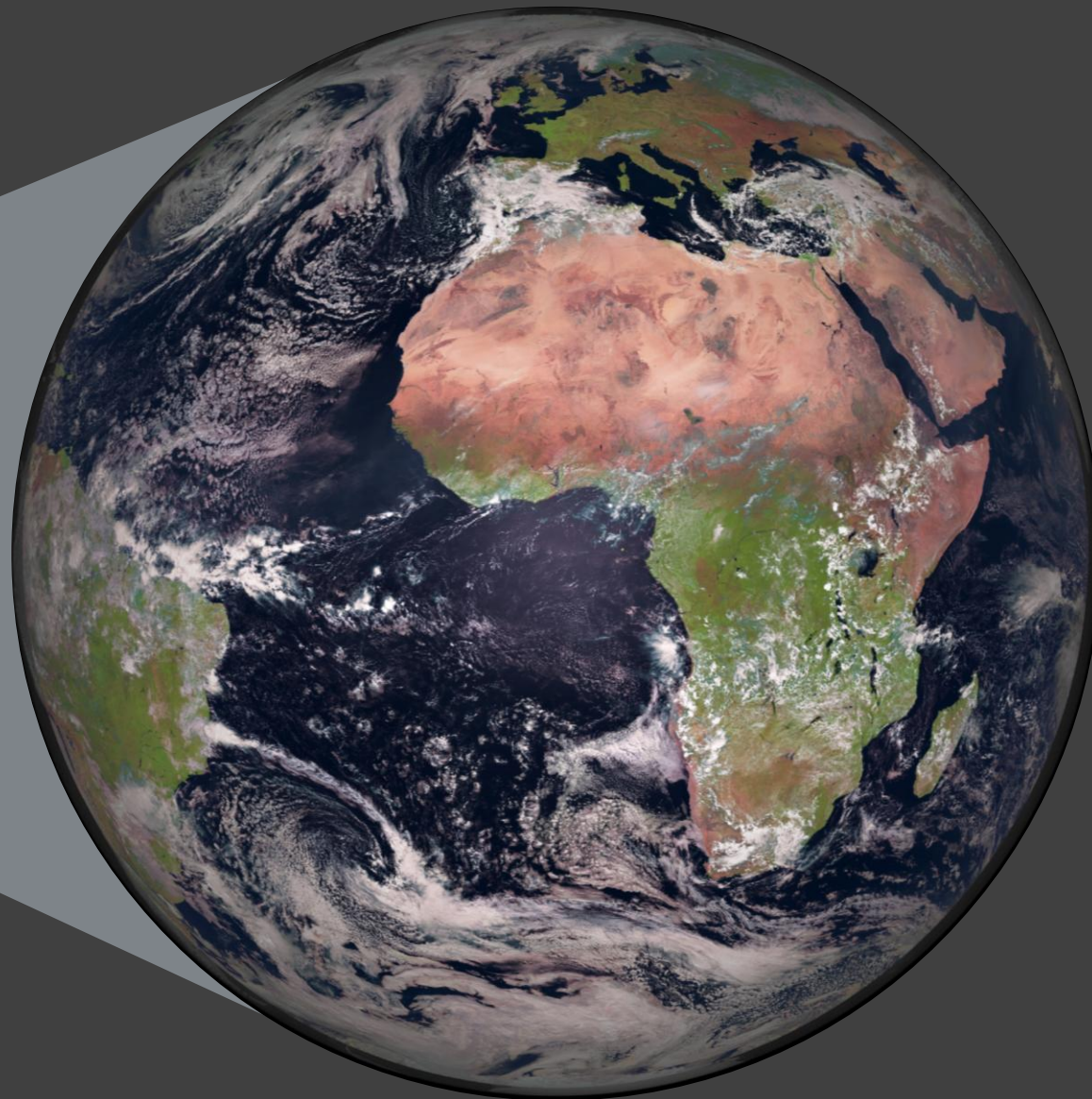


(+ vets, epidemiologists, climate scientists, public health, social scientists, microbiologists, geographers, evolutionary biologists...)

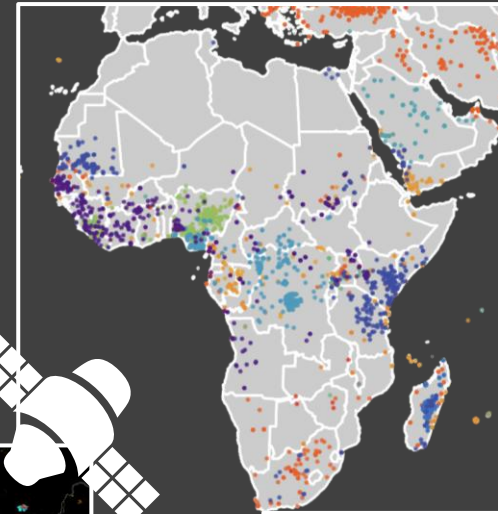
(Image: Sketchplanations)



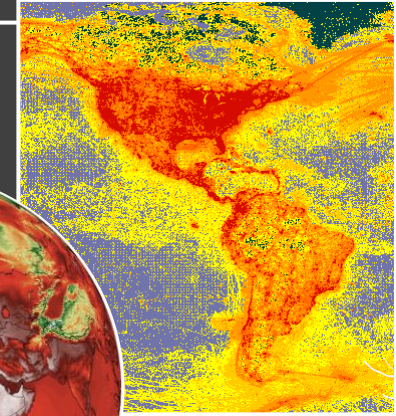
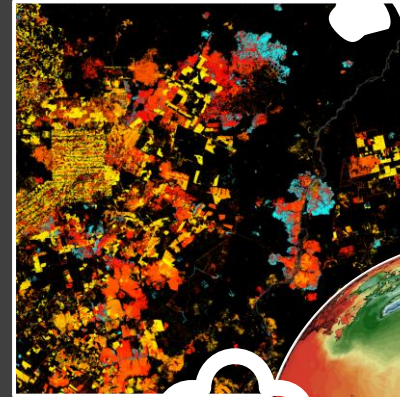
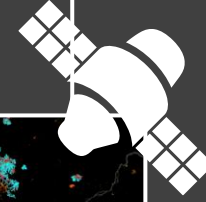
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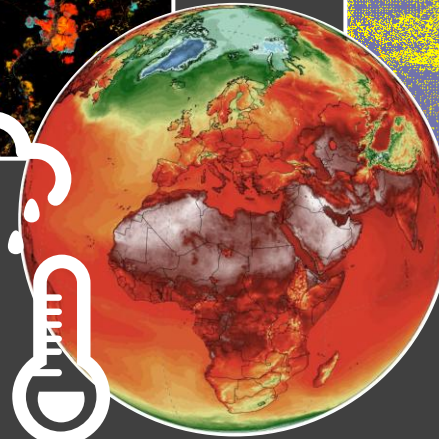
Global human
and animal
disease data



Earth
observation



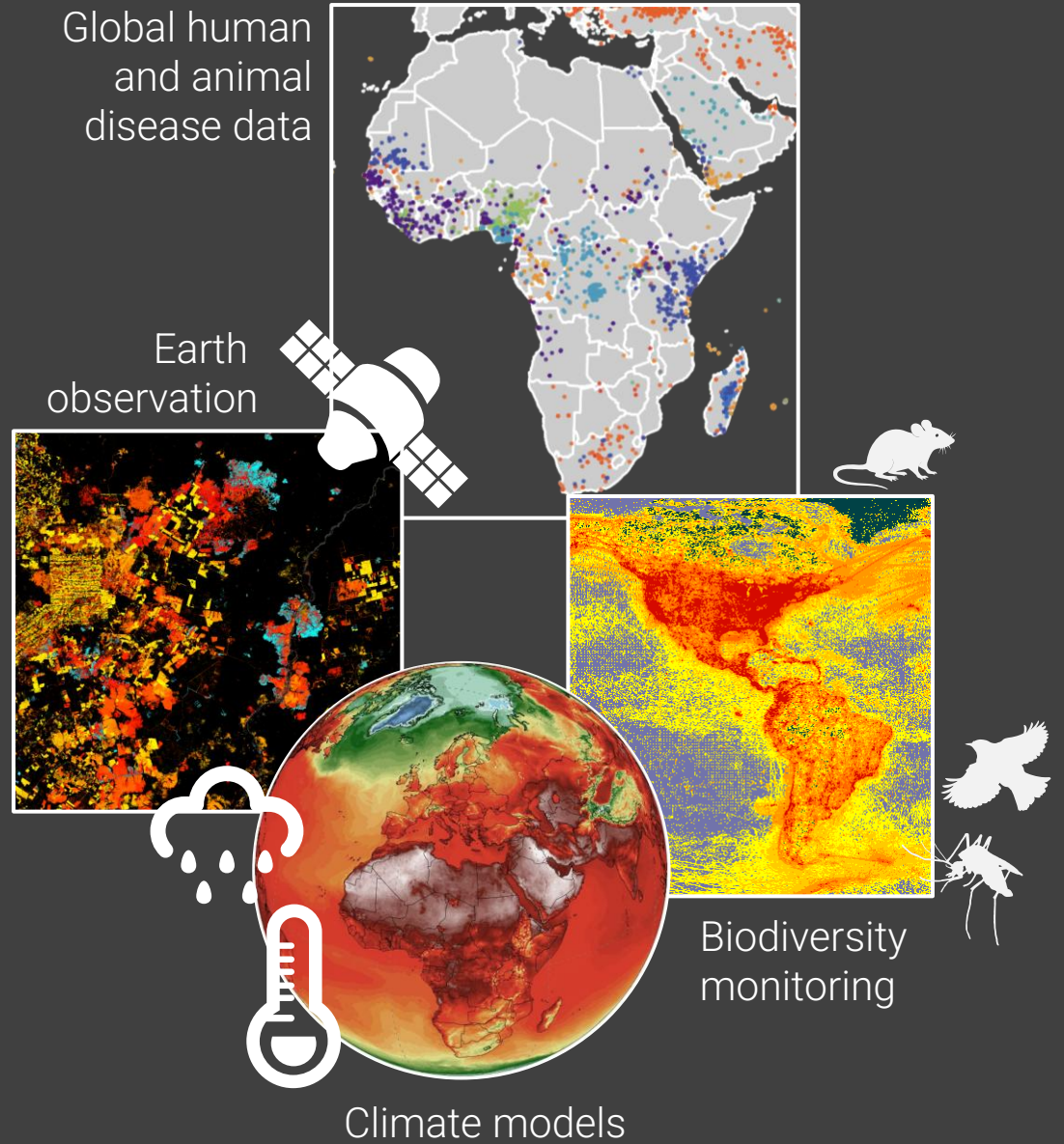
Biodiversity
monitoring



Climate models

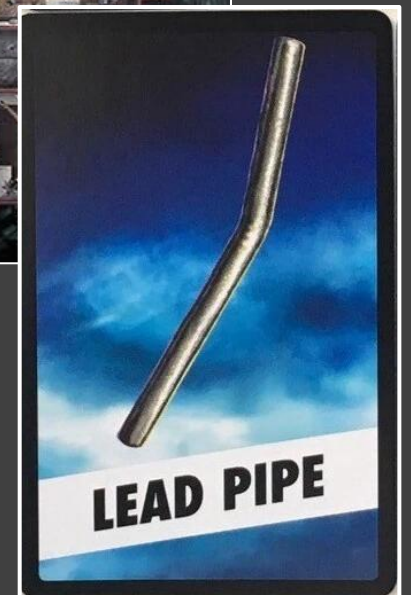
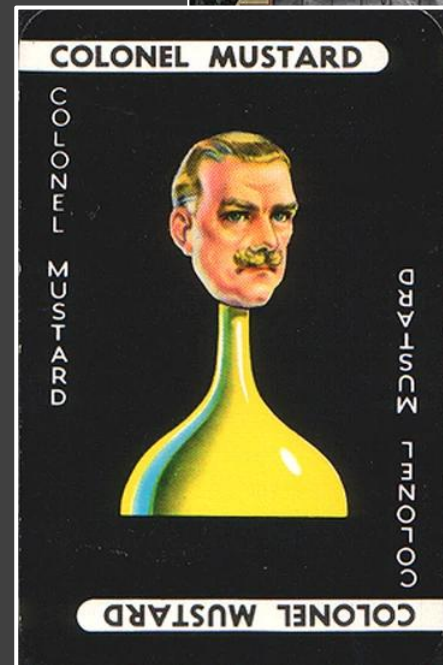


Searching for the fingerprint of
ecology and climate in large-
scale patterns of disease



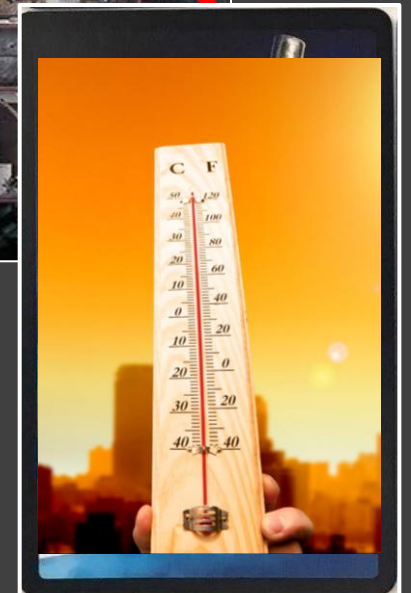
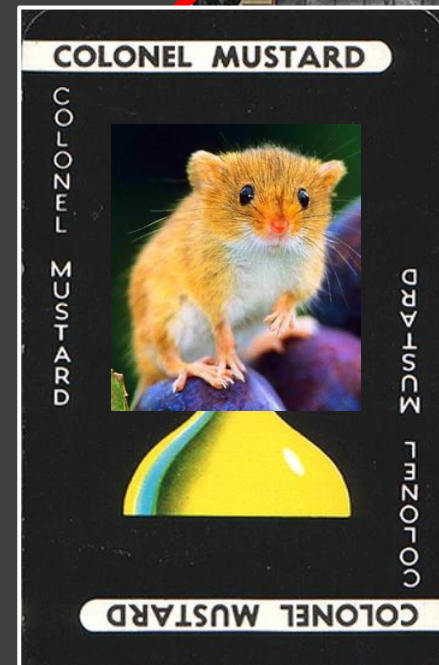
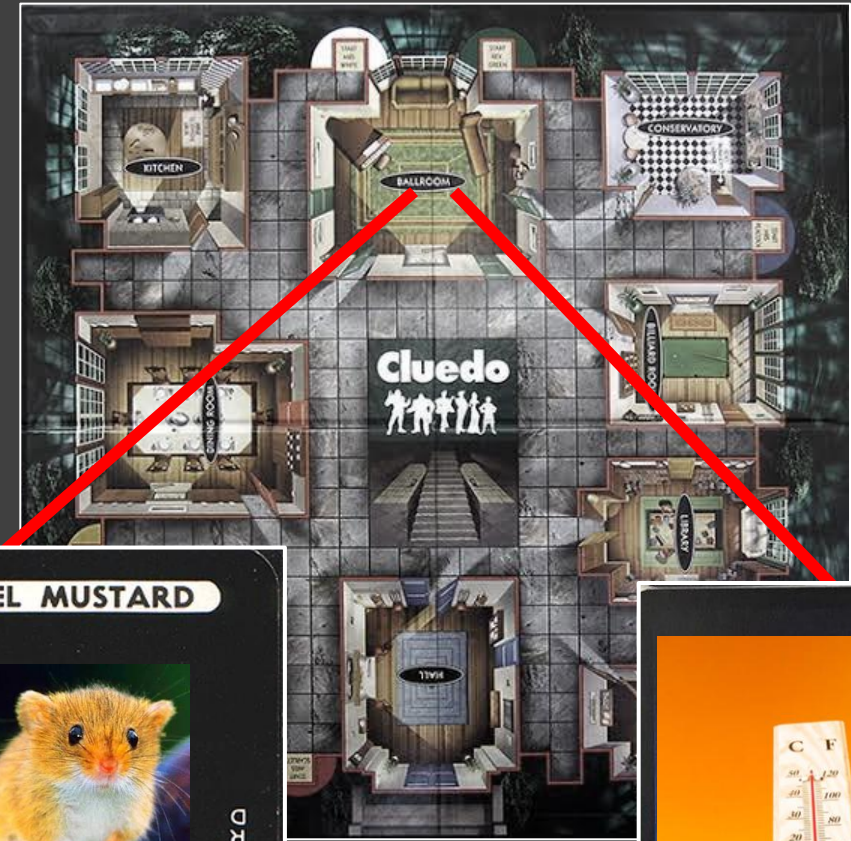


Searching for the fingerprint of ecology and climate in large-scale patterns of disease



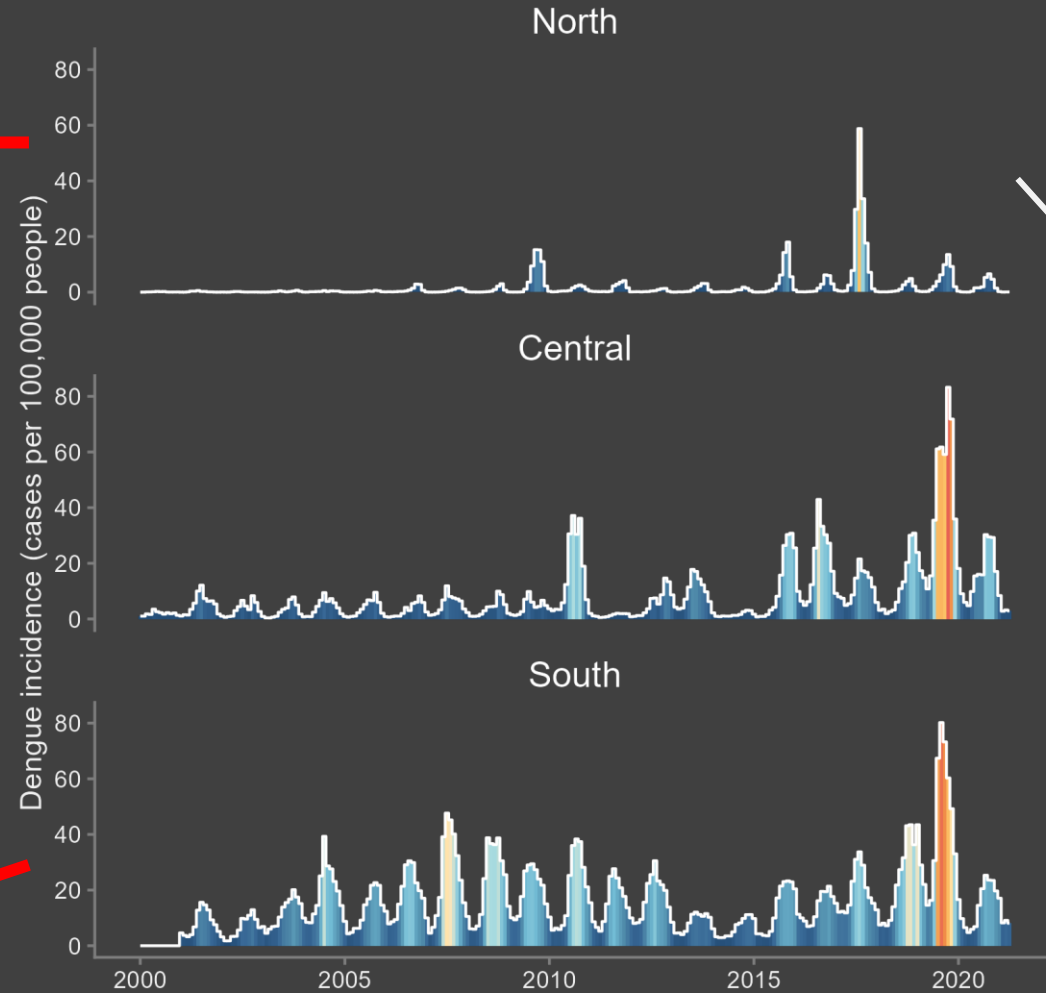
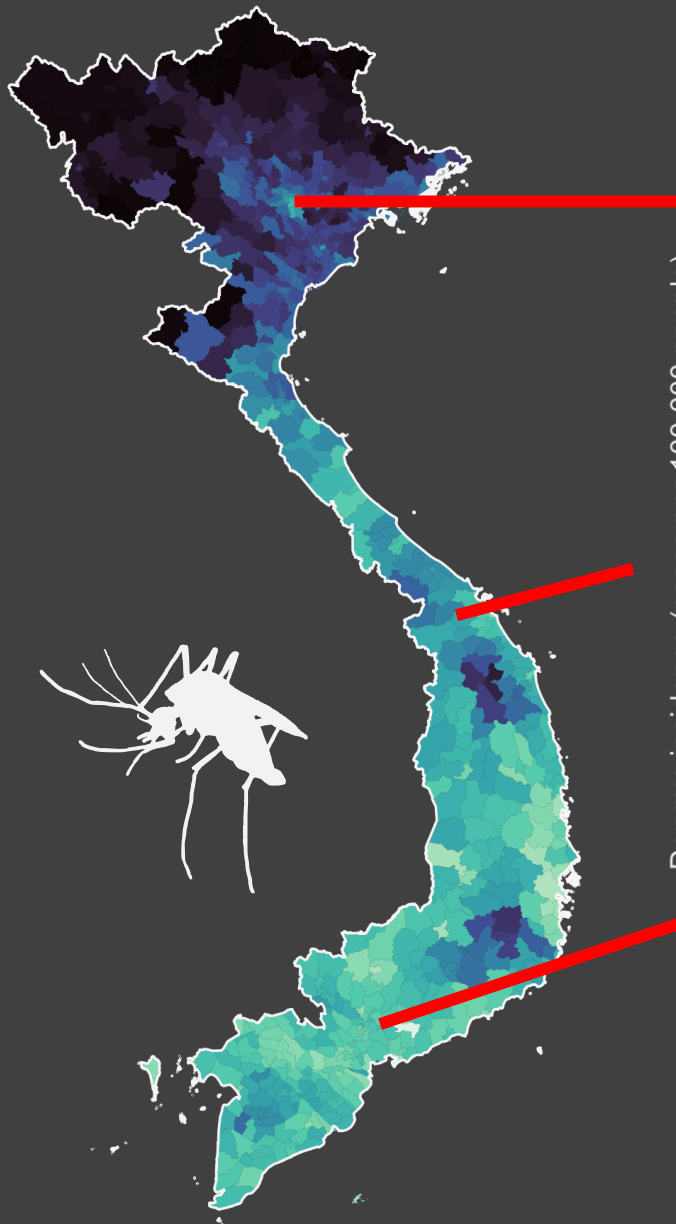


Searching for the fingerprint of ecology and climate in large-scale patterns of disease



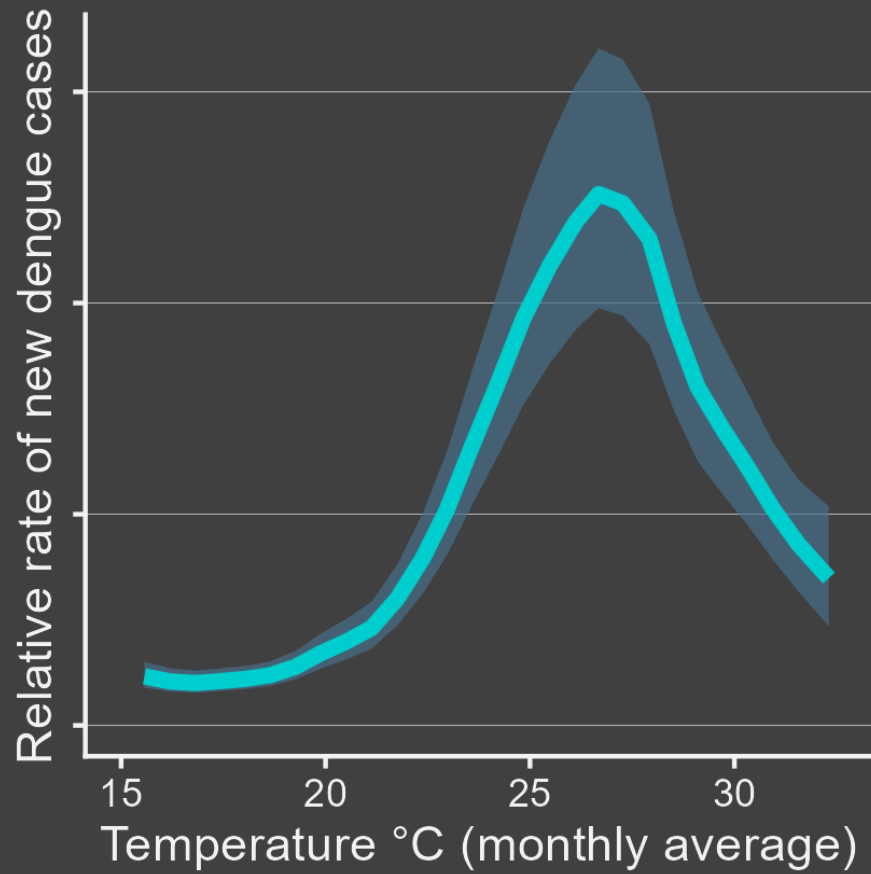
Is climate change contributing to the growth of dengue fever as a public health problem in Vietnam?





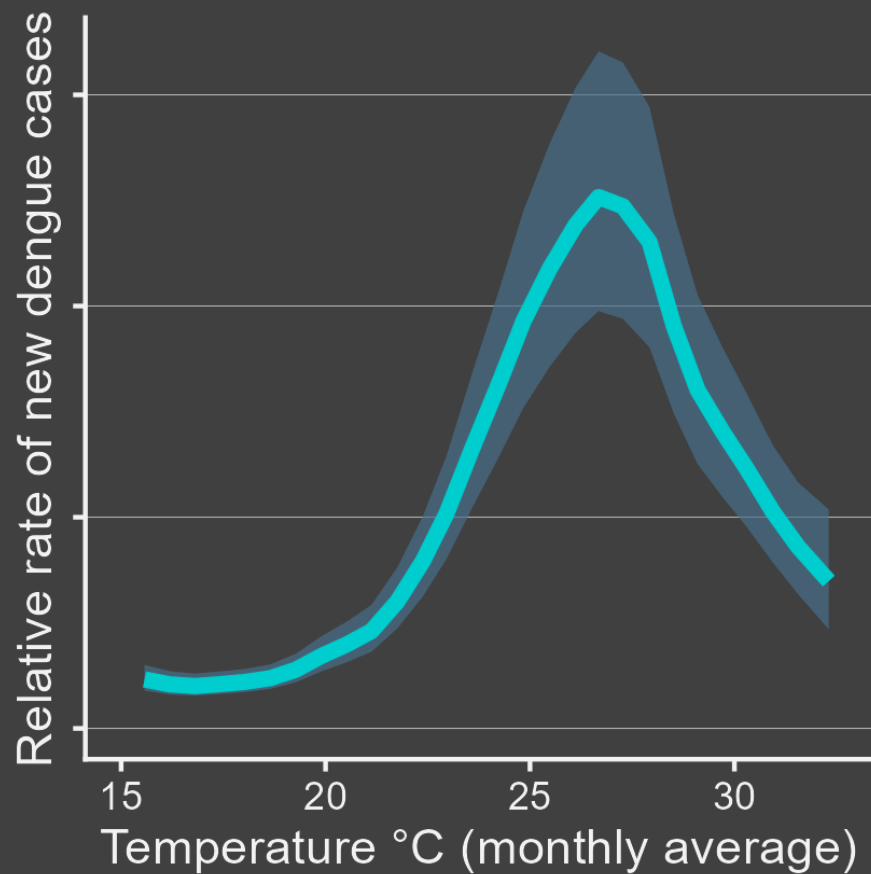
Statistical
models linking
disease
incidence with
climate and
ecosystem data

Monthly hospital reports of over 2 million
dengue fever cases since late 1990s

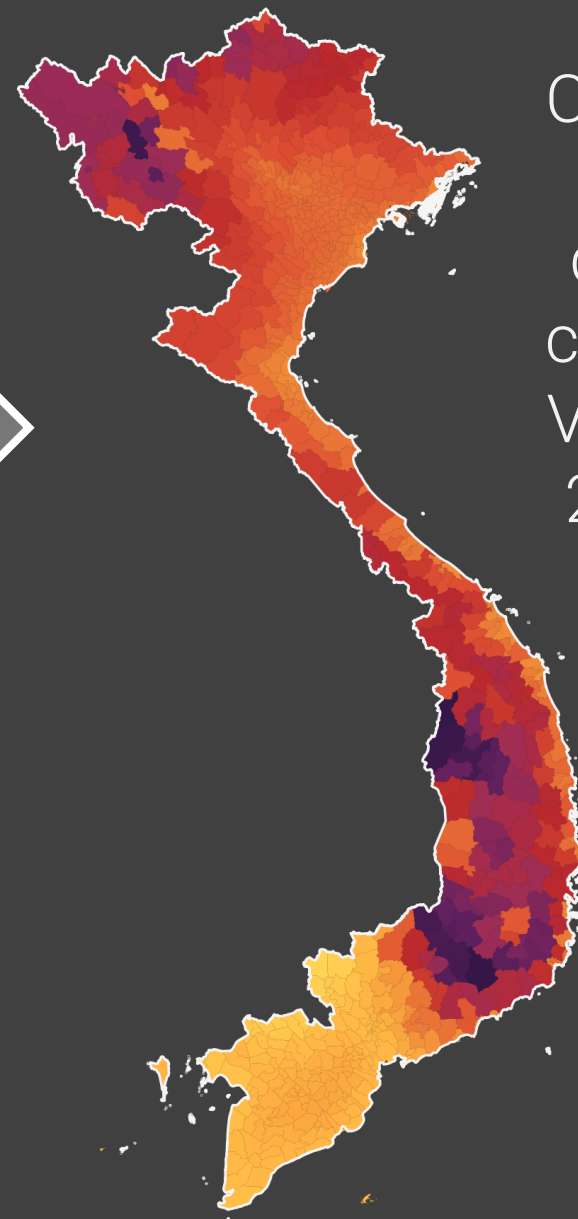


The rate of dengue spread depends on temperature— with the same shape that mosquito biology would predict!

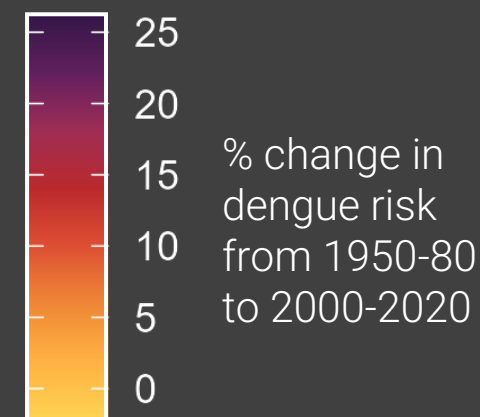
(Adapted from Gibb et al. 2023, Nature Communications)



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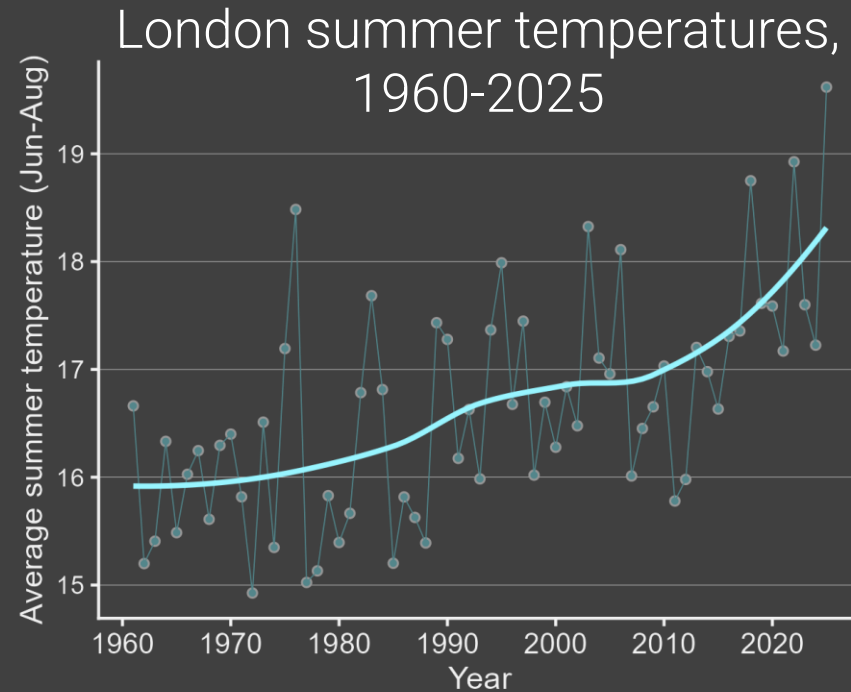


Climate warming has increased dengue rates in central and north Vietnam by up to 25% since 1950





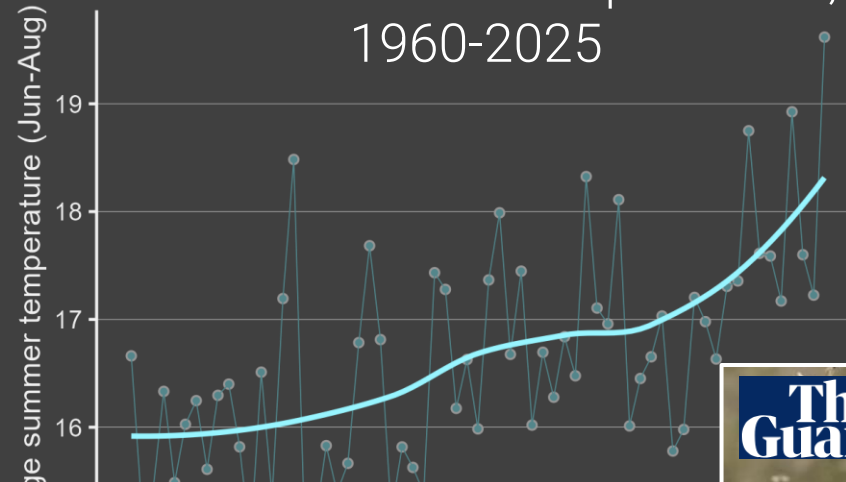
The arrival of mosquito-borne diseases in the UK is not a matter of if, but when



The arrival of mosquito-borne diseases in the UK is not a matter of if, but when



London summer temperatures, 1960-2025



Invasive tropical mosquito found breeding in east London

Climate crisis could increase incursions of *Aedes aegypti*, which can transmit Zika virus and dengue, experts say



First cases of mosquito-borne virus found in blackbirds in Scotland



BBC

The Guardian UK



West Nile virus detected in UK mosquitoes for first time

UKHSA says risk to public very low after detection in genetic material of mosquitoes caught in Nottinghamshire

Questions
so far?



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